

Conservation agriculture buffers persistent carbon loss under warming by microbial adaptation

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2026-06-04

Contents

0.1	Note	1
0.2	Software and Package Versions	1
0.3	Fig2 CUE, substrate traits, SUE and RAE	3
0.4	Fig3 Amplicon sequencing figures	18
0.5	Fig4 Metagenome analysis	32
0.6	Fig5 SEM analysis	57

0.1 Note

This R Markdown document was prepared by Ling Jun (), China Agricultural University, for the manuscript *Conservation agriculture buffers persistent carbon loss under warming by microbial adaptation*.

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```
rm(list = ls())
library(pacman)
p_load(
  openxlsx, tidyverse, cowplot, ggpubr, ggunchained, patchwork, otuSummary,
  lmerTest, lme4, nlme, ggh4x, vegan, rdacca.hp, ggtern, dplyr, ggplot2,
  Polychrome, car, emmeans, ggrepel, FactoMineR, factoextra, ggview,
  ggpmisc, piecewiseSEM, reshape2, MetBrewer, gRodon, emmeans, semEff
)

source("pipeline_code_upload.R")
```

0.2 Software and Package Versions

The analyses and figures were generated using the software environment recorded below.

```

software.version_df <- tibble::tibble(
  Software = c("R", "Platform", "Operating system", "Pandoc"),
  Version = c(
    paste0(R.version$major, ".", R.version$minor),
    R.version$platform,
    paste(Sys.info()[c("sysname", "release", "machine")], collapse = " "),
    as.character(rmarkdown::find_pandoc())$version)
)
knitr::kable/software.version_df, caption = "Software versions used for the analyses.")

```

Table 1: Software versions used for the analyses.

Software	Version
R	4.5.2
Platform	aarch64-apple-darwin20
Operating system	Darwin 25.5.0 arm64
Pandoc	3.6.3

```

package.version_df <- tibble::tibble(
  Package = sort(unique(c(
    "openxlsx", "tidyverse", "cowplot", "ggpubr", "ggunchained", "patchwork",
    "otuSummary", "lmerTest", "lme4", "nlme", "ggh4x", "vegan", "rdacca.hp",
    "ggtern", "dplyr", "ggplot2", "Polychrome", "car", "emmeans", "ggrepel",
    "FactoMineR", "factoextra", "ggview", "ggpmisc", "piecewiseSEM",
    "reshape2", "MetBrewer", "gRodon", "semEff", "pacman", "knitr",
    "rmarkdown"
  ))),
  Version = purrr::map_chr(Package, ~ as.character(utils::packageVersion(.x)))
)
knitr::kable(package.version_df, caption = "R package versions used for the analyses.")

```

Table 2: R package versions used for the analyses.

Package	Version
car	3.1.5
cowplot	1.2.0
dplyr	1.2.1
emmeans	2.0.2
factoextra	2.0.0
FactoMineR	2.13
ggh4x	0.3.1.9000
ggplot2	4.0.2
ggpmisc	0.7.0

Package	Version
ggpubr	0.6.3
ggrepel	0.9.8
ggtern	4.0.0
ggunchained	0.0.1
ggview	0.2.2
gRodon	2.7.2
knitr	1.51
lme4	2.0.1
lmerTest	3.2.1
MetBrewer	0.2.0
nlme	3.1.169
openxlsx	4.2.8.1
otuSummary	0.1.2
pacman	0.5.1
patchwork	1.3.2
piecewiseSEM	2.3.1
Polychrome	1.5.4
rdacca.hp	1.1.1
reshape2	1.4.5
rmarkdown	2.31
semEff	0.7.2
tidyverse	2.0.0
vegan	2.7.3

0.3 Fig2 CUE, substrate traits, SUE and RAE

```
df <- openxlsx::read.xlsx("data_for_RAE.xlsx", sheet = 1) %>%
  dplyr::mutate(Management = factor(Management, levels = c("Conventional", "Conservation")))

cue.summary_df <- df %>%
  dplyr::group_by(Treat) %>%
  dplyr::summarise(
    CUE = mean(CUE),
    SUE_Glu = mean(SUE_Glu),
    SUE_Van = mean(SUE_Van),
    .groups = "drop"
  ); head(cue.summary_df)

## # A tibble: 4 x 4
##   Treat      CUE SUE_Glu SUE_Van
##   <chr>   <dbl>   <dbl>   <dbl>
## 1 NN     0.160     0.730     0.580
## 2 NW     0.205     0.824     0.481
## 3 TN     0.181     0.750     0.710
```

```
## 4 TW      0.0460    0.854    0.743
```

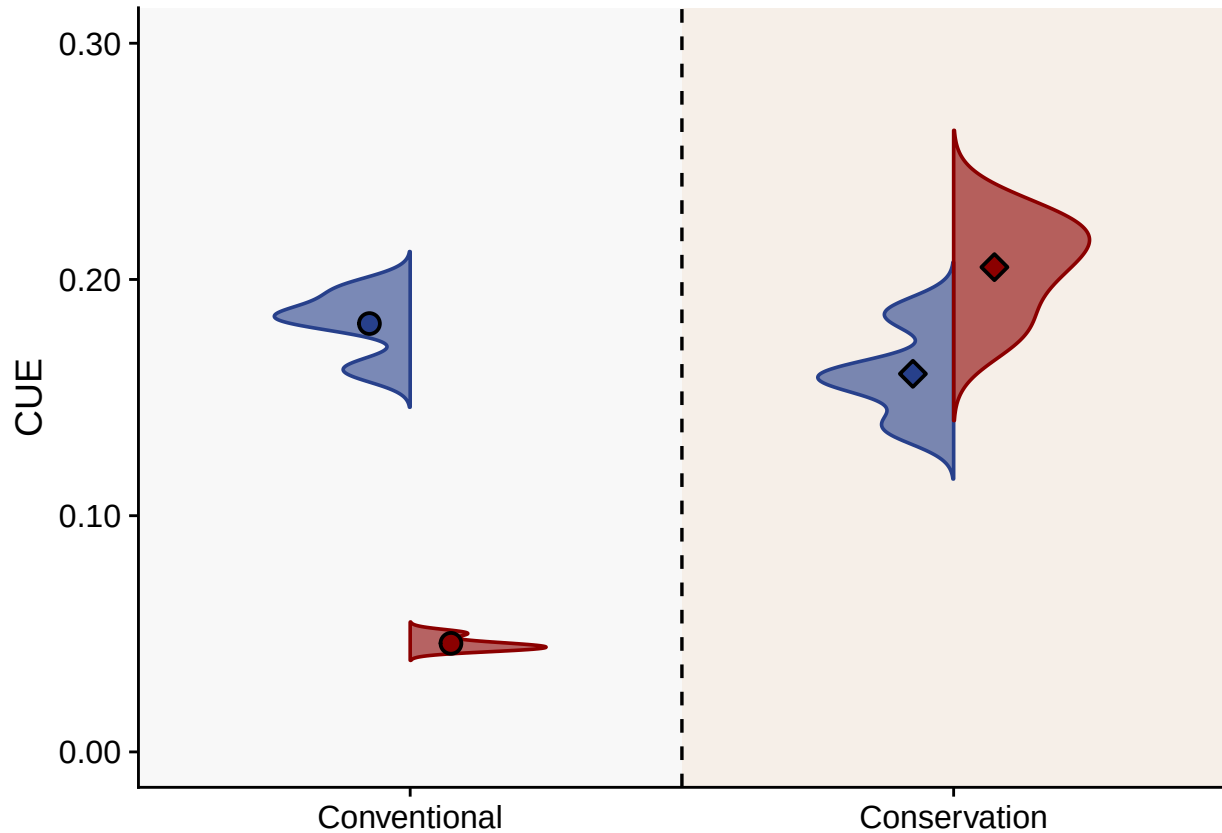
0.3.1 A. CUE

```
A.CUE_df <- df %>%  
  dplyr::select(Treat, Management, Warm, CUE) %>%  
  tidyr::pivot_longer(-c(Treat, Management, Warm), values_to = "value")  
head(A.CUE_df)
```

```
## # A tibble: 6 x 5  
##   Treat Management    Warm   name  value  
##   <chr> <fct>         <chr> <chr> <dbl>  
## 1 TN    Conventional Ambient CUE    0.196  
## 2 TN    Conventional Ambient CUE    0.185  
## 3 TN    Conventional Ambient CUE    0.162  
## 4 TN    Conventional Ambient CUE    0.183  
## 5 TW    Conventional Warming CUE    0.0503  
## 6 TW    Conventional Warming CUE    0.0435
```

```
A.CUE_plot <- ggplot(A.CUE_df, aes(x = Management, y = value, fill = Warm, color = Warm, shape = Warm)) +  
  geom_rect(  
    data = background_data,  
    aes(xmin = xmin, xmax = xmax, ymin = -Inf, ymax = Inf),  
    fill = background_data$fill_color,  
    alpha = 0.1,  
    inherit.aes = FALSE  
  ) +  
  ggunchained::geom_split_violin(alpha = 0.6, trim = FALSE, width = 0.5, scale = "width") +  
  stat_summary(  
    fun = "mean",  
    geom = "point",  
    position = position_dodge(0.3),  
    show.legend = FALSE,  
    size = 3,  
    color = "black",  
    stroke = 1  
  ) +  
  geom_vline(xintercept = 1.5, linetype = "dashed", color = "black", linewidth = 0.6) +  
  scale_fill_manual(values = cols1) +  
  scale_color_manual(values = cols1) +  
  scale_shape_manual(values = c(21, 23)) +  
  scale_y_continuous(labels = scales::number_format(accuracy = 0.01)) +  
  scale_x_discrete(expand = expansion(add = 0.2)) +  
  coord_cartesian(ylim = c(0, 0.3)) +  
  labs(y = "CUE", x = NULL, color = NULL, fill = NULL) +
```

```
my_theme +
  theme(legend.position = "none", axis.title.x = element_blank())
A.CUE_plot
```



0.3.2 B and C. Substrate traits

```
B.substrate_df <- df %>%
  dplyr::select(Treat, Management, Warm, Block, LCI, Root_exudation, DOC_SOC, MAOC_POC)
head(B.substrate_df)
```

##	Treat	Management	Warm	Block	LCI	Root_exudation	DOC_SOC	MAOC_POC
## 1	TN	Conventional	Ambient	B1	0.1409314	3.340918	6.299706	1.410428
## 2	TN	Conventional	Ambient	B2	0.1439454	2.825028	6.179547	1.248514
## 3	TN	Conventional	Ambient	B3	0.1459575	3.160674	6.134653	1.214896
## 4	TN	Conventional	Ambient	B4	0.1411711	4.087253	6.199054	1.285984
## 5	TW	Conventional	Warming	B1	0.1589486	4.822374	6.671324	1.653365
## 6	TW	Conventional	Warming	B2	0.1610468	4.883064	7.174591	1.795468

```

B.substrate.summary_df <- B.substrate_df %>%
  dplyr::group_by(Treat) %>%
  dplyr::summarise(
    dplyr::across(c(LCI, Root_exudation, DOC_SOC, MAOC_POC), mean),
    .groups = "drop"
  )
head(B.substrate.summary_df)

```

```

## # A tibble: 4 x 5
##   Treat    LCI Root_exudation DOC_SOC MAOC_POC
##   <chr> <dbl>         <dbl>   <dbl>   <dbl>
## 1 NN    0.160           3.84    6.60    1.17
## 2 NW    0.145           5.69    7.15    1.16
## 3 TN    0.143           3.35    6.20    1.29
## 4 TW    0.158           4.62    6.90    1.74

```

```

B.substrate.scaled_df <- B.substrate_df %>%
  dplyr::select(Root_exudation, DOC_SOC, MAOC_POC, LCI) %>%
  scale() %>%
  as.data.frame()
head(B.substrate.scaled_df)

```

```

##   Root_exudation  DOC_SOC  MAOC_POC      LCI
## 1   -1.0452496 -1.0661473  0.2784433 -1.2499046
## 2   -1.5662625 -1.3773846 -0.3564475 -0.8875942
## 3   -1.2272836 -1.4936694 -0.4882664 -0.6457204
## 4   -0.2915019 -1.3268583 -0.2095224 -1.2210825
## 5    0.4509197 -0.1035781  1.2310353  0.9159079
## 6    0.5122126  1.1999933  1.7882431  1.1681285

```

```

B.pca_result <- vegan::rda(B.substrate.scaled_df)
B.pca.importance_df <- summary(B.pca_result)$cont$importance %>%
  as.data.frame()
head(B.pca.importance_df)

```

```

##              PC1      PC2      PC3      PC4
## Eigenvalue      1.8533018 1.3494121 0.6847999 0.11248618
## Proportion Explained 0.4633254 0.3373530 0.1712000 0.02812155
## Cumulative Proportion 0.4633254 0.8006785 0.9718785 1.00000000

```

```

B.pca.sites_df <- vegan::scores(B.pca_result, display = "sites") %>%
  as.data.frame() %>%
  dplyr::mutate(
    Temperature = B.substrate_df$Warm,
    System = B.substrate_df$Management,
  )

```

```

    Treat = B.substrate_df$Treat
  )
head(B.pca.sites_df)

```

```

##           PC1           PC2 Temperature      System Treat
## 1 -0.7973000 -0.2859145      Ambient Conventional    TN
## 2 -1.1452315 -0.2997329      Ambient Conventional    TN
## 3 -1.0692650 -0.2970878      Ambient Conventional    TN
## 4 -0.6673199 -0.5848288      Ambient Conventional    TN
## 5  0.2736783  0.8322483      Warming Conventional    TW
## 6  0.8433560  1.1604410      Warming Conventional    TW

```

```

B.pca.vars_df <- vegan::scores(B.pca_result, display = "species") %>%
  as.data.frame() %>%
  tibble::rownames_to_column("varname")
head(B.pca.vars_df)

```

```

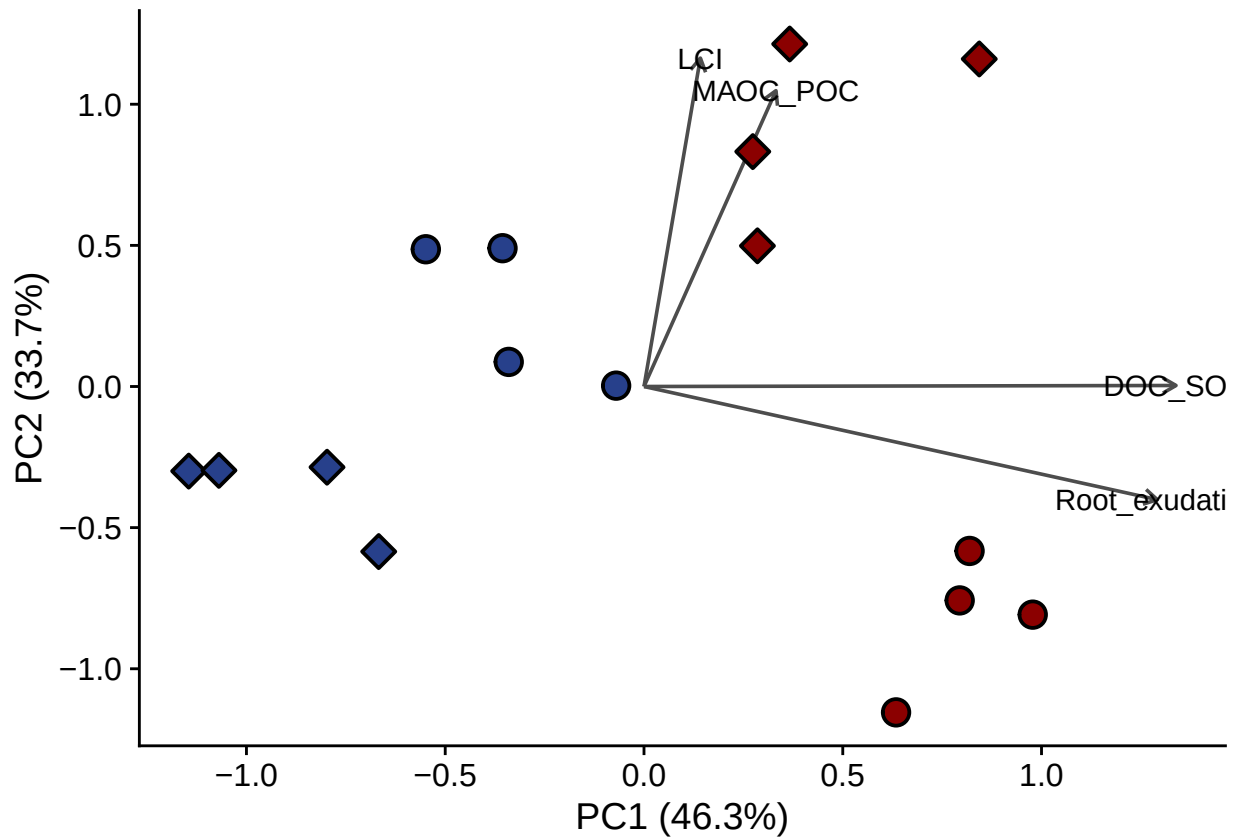
##           varname           PC1           PC2
## 1 Root_exudation 1.2912877 -0.400983318
## 2          DOC_SOC 1.3383440  0.003603242
## 3         MAOC_POC 0.3319619  1.048138158
## 4              LCI 0.1418312  1.163499543

```

```

B.substrate.pca_plot <- ggplot(B.pca.sites_df, aes(x = PC1, y = PC2)) +
  geom_segment(
    data = B.pca.vars_df,
    aes(x = 0, y = 0, xend = PC1, yend = PC2),
    arrow = arrow(length = unit(0.2, "cm")),
    color = "gray30"
  ) +
  geom_text(data = B.pca.vars_df, aes(x = PC1, y = PC2, label = varname), size = 4) +
  geom_point(aes(shape = System, fill = Temperature), size = 4, alpha = 1, stroke = 1) +
  scale_fill_manual(values = cols1) +
  scale_shape_manual(values = c(23, 21)) +
  labs(
    x = paste0("PC1 (", round(B.pca.importance_df[2, 1] * 100, 1), "%)"),
    y = paste0("PC2 (", round(B.pca.importance_df[2, 2] * 100, 1), "%)"),
  ) +
  my_theme +
  theme(legend.position = "none"); B.substrate.pca_plot

```



```
C.pca.var_df <- FactoMineR::PCA(B.substrate.scaled_df, graph = FALSE)$var$contrib %>%
  as.data.frame() %>%
  tibble::rownames_to_column("rowname")
head(C.pca.var_df)
```

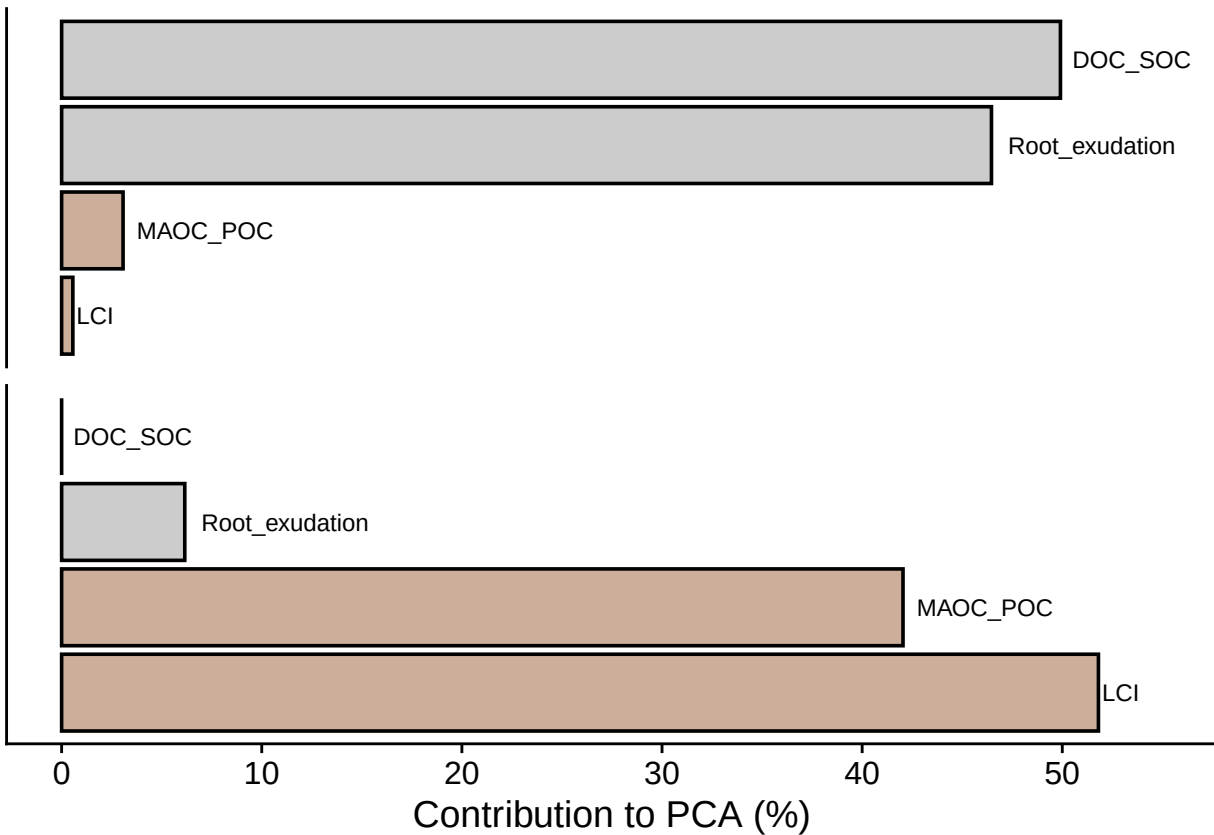
```
##           rowname      Dim.1      Dim.2      Dim.3      Dim.4
## 1 Root_exudation 46.4605408 6.153077e+00 0.4564739 46.92990812
## 2          DOC_SQC 49.9084116 4.968514e-04 3.2664348 46.82465680
## 3          MAOC_POC 3.0705399 4.204137e+01 54.8721470 0.01594755
## 4             LCI 0.5605077 5.180506e+01 41.4049443 6.22948753
```

```
C.pca.contribution_df <- C.pca.var_df %>%
  dplyr::mutate(group = c("access", "access", "quality", "quality")) %>%
  dplyr::select(rowname, group, Dim.1, Dim.2) %>%
  tidyr::pivot_longer(-c(rowname, group)) %>%
  dplyr::group_by(name) %>%
  dplyr::mutate(rowname_ordered = factor(rowname, levels = rowname[order(value)])) %>%
  dplyr::ungroup()
head(C.pca.contribution_df)
```

```
## # A tibble: 6 x 5
```


##	rowname	group	name	value	rowname_ordered
##	<chr>	<chr>	<chr>	<dbl>	<fct>
## 1	Root_exudation	access	Dim.1	46.5	Root_exudation
## 2	Root_exudation	access	Dim.2	6.15	Root_exudation
## 3	DOC_SOC	access	Dim.1	49.9	DOC_SOC
## 4	DOC_SOC	access	Dim.2	0.000497	DOC_SOC
## 5	MAOC_POC	quality	Dim.1	3.07	MAOC_POC
## 6	MAOC_POC	quality	Dim.2	42.0	MAOC_POC

```
C.pca.contribution_plot <- ggplot(C.pca.contribution_df, aes(x = rowname_ordered, y = value, f
  geom_bar(stat = "identity", color = "black", alpha = 1) +
  geom_text(aes(label = rowname), size = 3, hjust = -0.1) +
  coord_flip() +
  facet_wrap(~name, ncol = 1) +
  scale_y_continuous(limits = c(0, 55), breaks = seq(0, 50, 10)) +
  scale_fill_manual(values = rev(c("#ccaf9a", "grey80"))) +
  ylab("Contribution to PCA (%)") +
  my_theme +
  theme(
    panel.background = element_blank(),
    legend.position = "none",
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank(),
    axis.title.y = element_blank(),
    axis.line.x = element_line(),
    axis.line.y = element_line(),
    plot.title = element_blank(),
    strip.text = element_blank()
  ); C.pca.contribution_plot
```



0.3.3 D and E. SUE

```
D.SUE.Glu_df <- df %>%
  dplyr::select(Treat, Management, Warm, SUE_Glu) %>%
  tidyr::pivot_longer(-c(Treat, Management, Warm), values_to = "value")
head(D.SUE.Glu_df)
```

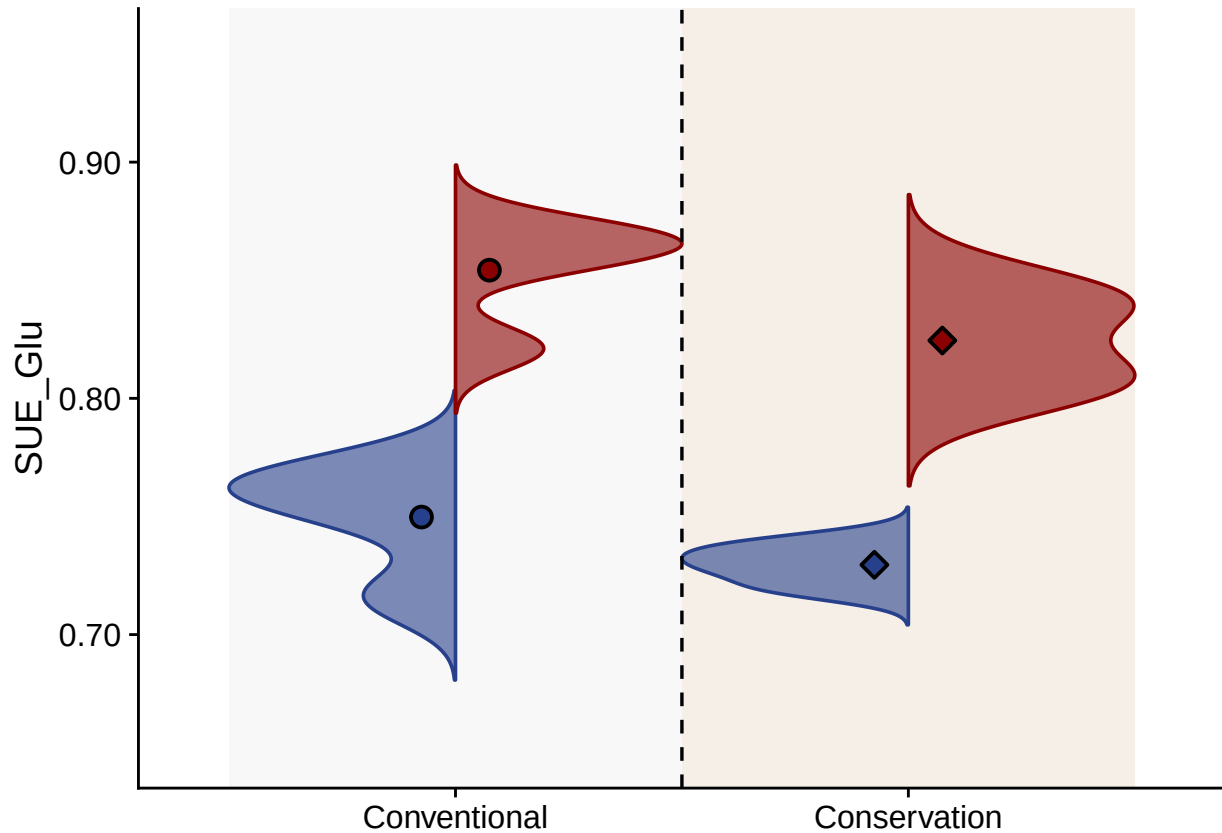
```
## # A tibble: 6 x 5
##   Treat Management    Warm   name   value
##   <chr> <fct>         <chr> <chr> <dbl>
## 1 TN    Conventional Ambient SUE_Glu 0.716
## 2 TN    Conventional Ambient SUE_Glu 0.751
## 3 TN    Conventional Ambient SUE_Glu 0.768
## 4 TN    Conventional Ambient SUE_Glu 0.764
## 5 TW    Conventional Warming SUE_Glu 0.871
## 6 TW    Conventional Warming SUE_Glu 0.859
```

```
D.SUE.Glu_plot <- ggplot(D.SUE.Glu_df, aes(x = Management, y = value, fill = Warm, color = Warm))
  geom_rect(
    data = background_data,
```

```

    aes(xmin = xmin, xmax = xmax, ymin = -Inf, ymax = Inf),
    fill = background_data$fill_color,
    alpha = 0.1,
    inherit.aes = FALSE
) +
ggunchained::geom_split_violin(alpha = 0.6, trim = FALSE, width = 1, scale = "width") +
stat_summary(
  fun = "mean",
  geom = "point",
  position = position_dodge(0.3),
  show.legend = FALSE,
  size = 3,
  color = "black",
  stroke = 1
) +
geom_vline(xintercept = 1.5, linetype = "dashed", color = "black", linewidth = 0.6) +
scale_fill_manual(values = cols1) +
scale_color_manual(values = cols1) +
scale_shape_manual(values = c(21, 23)) +
scale_y_continuous(labels = scales::number_format(accuracy = 0.01)) +
scale_x_discrete(expand = expansion(add = 0.7)) +
coord_cartesian(ylim = c(0.65, 0.95)) +
labs(y = "SUE_Glu", x = NULL, color = NULL, fill = NULL) +
my_theme +
theme(legend.position = "none", axis.title.x = element_blank())
D.SUE.Glu_plot

```



```
E.SUE.Van_df <- df %>%
  dplyr::select(Treat, Management, Warm, SUE_Van) %>%
  tidyr::pivot_longer(-c(Treat, Management, Warm), values_to = "value")
head(E.SUE.Van_df)
```

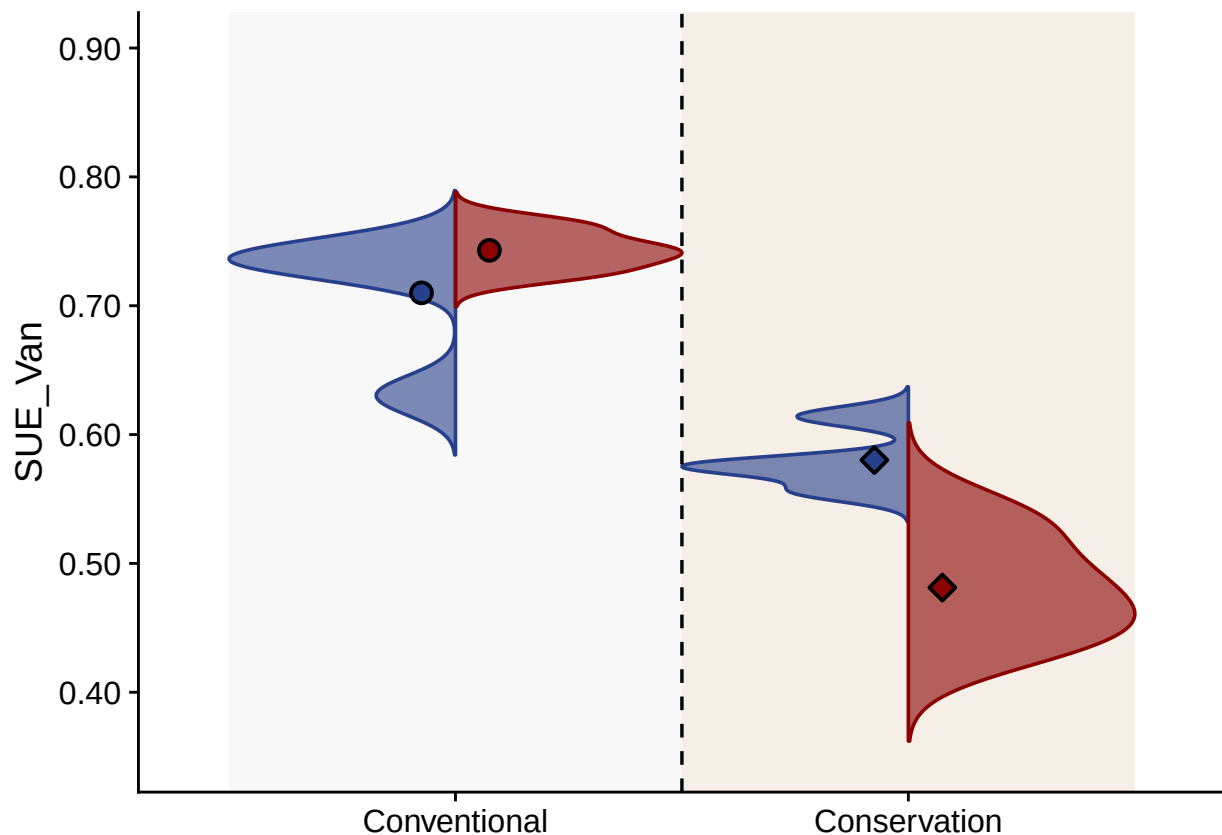
```
## # A tibble: 6 x 5
##   Treat Management    Warm   name   value
##   <chr> <fct>         <chr> <chr> <dbl>
## 1 TN    Conventional Ambient SUE_Van 0.743
## 2 TN    Conventional Ambient SUE_Van 0.732
## 3 TN    Conventional Ambient SUE_Van 0.735
## 4 TN    Conventional Ambient SUE_Van 0.630
## 5 TW    Conventional Warming  SUE_Van 0.763
## 6 TW    Conventional Warming  SUE_Van 0.747
```

```
E.SUE.Van_plot <- ggplot(E.SUE.Van_df, aes(x = Management, y = value, fill = Warm, color = Warm)) +
  geom_rect(
    data = background_data,
    aes(xmin = xmin, xmax = xmax, ymin = -Inf, ymax = Inf),
    fill = background_data$fill_color,
    alpha = 0.1,
    inherit.aes = FALSE
```

```

) +
ggunchained::geom_split_violin(alpha = 0.6, trim = FALSE, width = 1, scale = "width") +
stat_summary(
  fun = "mean",
  geom = "point",
  position = position_dodge(0.3),
  show.legend = FALSE,
  size = 3,
  color = "black",
  stroke = 1
) +
geom_vline(xintercept = 1.5, linetype = "dashed", color = "black", linewidth = 0.6) +
scale_fill_manual(values = cols1) +
scale_color_manual(values = cols1) +
scale_shape_manual(values = c(21, 23)) +
scale_y_continuous(limits = c(0.35, 0.9), labels = scales::number_format(accuracy = 0.01)) +
scale_x_discrete(expand = expansion(add = 0.7)) +
labs(y = "SUE_Van", x = NULL, color = NULL, fill = NULL) +
my_theme +
theme(legend.position = "none")
E.SUE.Van_plot

```



0.3.4 F. RAE

```
F.growth.resp_df <- df %>%
  dplyr::select(Treat, Block, Management, Warm, CUE_G, CUE_R, SUE_Glu_G, SUE_Glu_R, SUE_Van_G,
  tidyr::pivot_longer(
    cols = dplyr::matches("_ (G|R)$"),
    names_to = c("group", "flux"),
    names_pattern = "(.+)_ (G|R)$",
    values_to = "value"
  ) %>%
  dplyr::group_by(Treat, Management, Warm, group, flux) %>%
  dplyr::summarise(mean_value = mean(value), .groups = "drop") %>%
  tidyr::pivot_wider(names_from = flux, values_from = mean_value)
head(F.growth.resp_df)
```

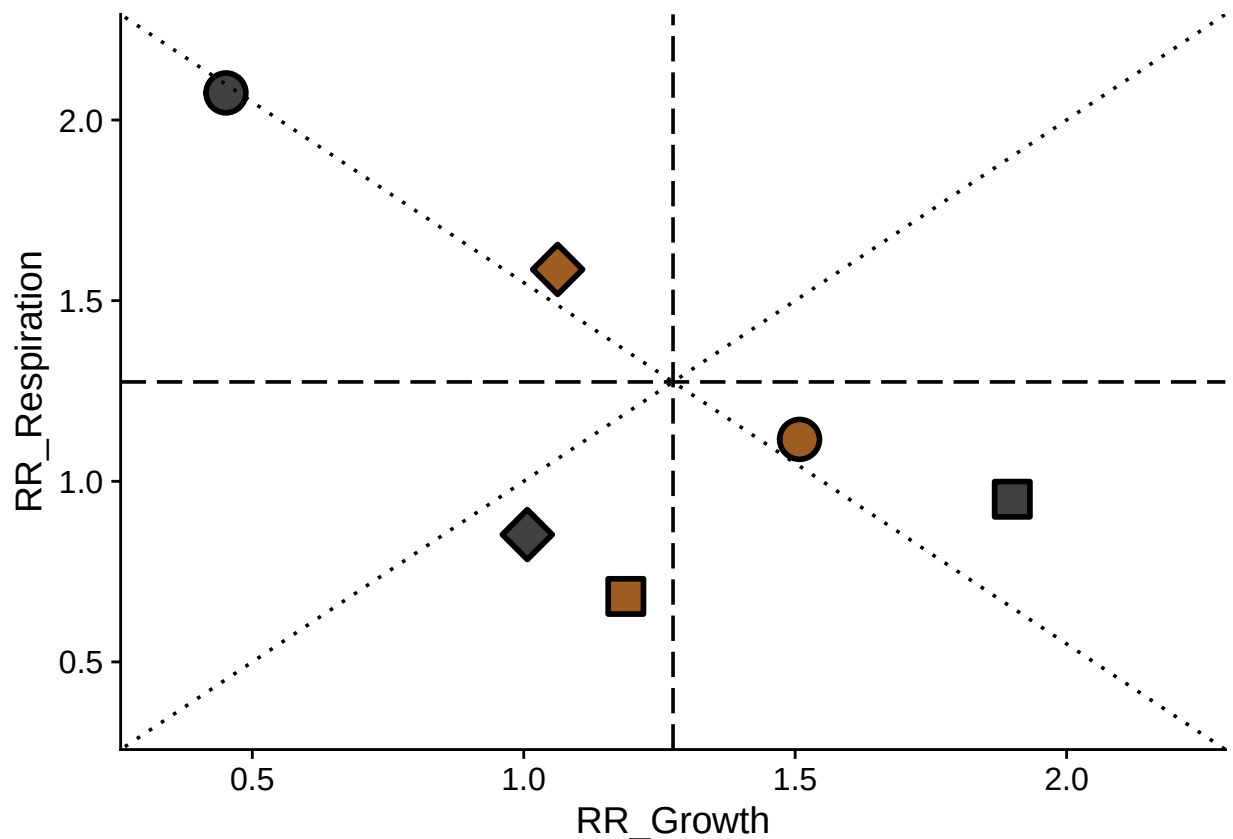
```
## # A tibble: 6 x 6
##   Treat Management   Warm   group      G      R
##   <chr> <fct>         <chr> <chr>    <dbl> <dbl>
## 1 NN    Conservation Ambient CUE      123.  646.
## 2 NN    Conservation Ambient SUE_Glu   2.07   0.769
## 3 NN    Conservation Ambient SUE_Van   0.491  0.355
## 4 NW    Conservation Warming  CUE      186.  721.
## 5 NW    Conservation Warming  SUE_Glu   2.46   0.524
## 6 NW    Conservation Warming  SUE_Van   0.521  0.563
```

```
F.warming.ratio_df <- F.growth.resp_df %>%
  tidyr::pivot_wider(
    id_cols = c(Management, group),
    names_from = Warm,
    values_from = c(G, R),
    names_sep = "_"
  ) %>%
  dplyr::mutate(
    mean_G_ratio = G_Warming / G_Ambient,
    mean_R_ratio = R_Warming / R_Ambient
  ) %>%
  dplyr::select(Management, group, mean_G_ratio, mean_R_ratio)
head(F.warming.ratio_df)
```

```
## # A tibble: 6 x 4
##   Management   group mean_G_ratio mean_R_ratio
##   <fct>      <chr>      <dbl>      <dbl>
## 1 Conservation CUE          1.51        1.12
## 2 Conservation SUE_Glu       1.19        0.681
## 3 Conservation SUE_Van       1.06        1.59
```

## 4 Conventional CUE	0.451	2.08
## 5 Conventional SUE_Glu	1.90	0.950
## 6 Conventional SUE_Van	1.01	0.853

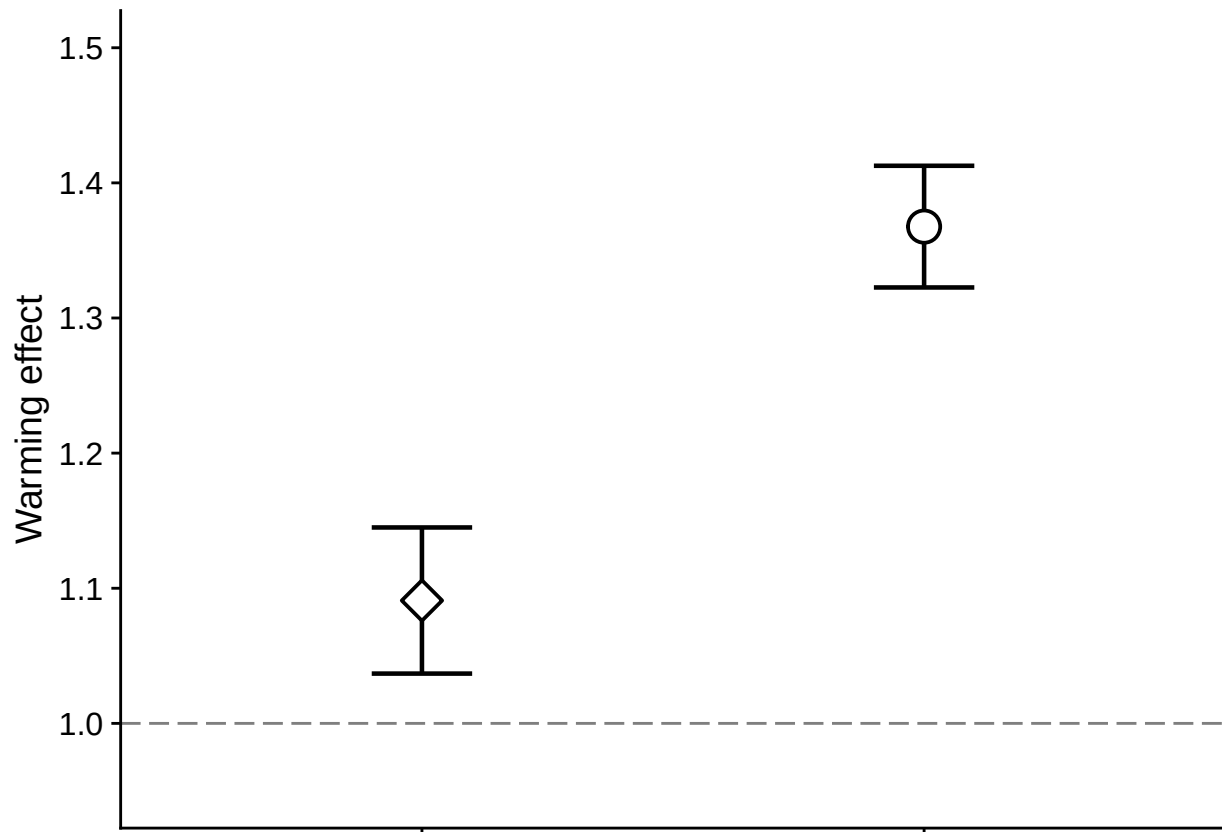
```
F.G.R_plot <- ggplot(
  F.warming.ratio_df,
  aes(x = mean_G_ratio, y = mean_R_ratio, shape = group, fill = Management)
) +
  geom_point(size = 6, stroke = 1.5) +
  geom_hline(yintercept = 1.275, linetype = "longdash", color = "black") +
  geom_vline(xintercept = 1.275, linetype = "longdash", color = "black") +
  geom_abline(intercept = 0, slope = 1, linetype = "dotted", color = "black") +
  geom_abline(intercept = 2.55, slope = -1, linetype = "dotted", color = "black") +
  scale_shape_manual(values = c(21, 22, 23)) +
  scale_fill_manual(values = c("#404040", "#9d5d21")) +
  scale_x_continuous(limits = c(0.35, 2.2)) +
  scale_y_continuous(limits = c(0.35, 2.2)) +
  labs(x = "RR_Growth", y = "RR_Respiration") +
  my_theme +
  theme(legend.position = "none")
F.G.R_plot
```



```
F.RAE_df <- df %>%
  dplyr::select(Treat, Management, Block, Warm, SUE_Glu, SUE_Van) %>%
  dplyr::mutate(prefer = SUE_Glu / SUE_Van) %>%
  dplyr::select(-SUE_Glu, -SUE_Van) %>%
  tidyr::pivot_longer(-c(Treat, Management, Warm, Block))
head(F.RAE_df)
```

```
## # A tibble: 6 x 6
##   Treat Management   Block Warm   name   value
##   <chr> <fct>         <chr> <chr>  <chr>  <dbl>
## 1 TN    Conventional B1    Ambient prefer 0.963
## 2 TN    Conventional B2    Ambient prefer 1.03
## 3 TN    Conventional B3    Ambient prefer 1.05
## 4 TN    Conventional B4    Ambient prefer 1.21
## 5 TW    Conventional B1    Warming prefer 1.14
## 6 TW    Conventional B2    Warming prefer 1.15
```

```
F.RAE_plot <- df %>%
  dplyr::select(Treat, Management, Block, Warm, SUE_Glu, SUE_Van) %>%
  dplyr::mutate(prefer = SUE_Glu / SUE_Van) %>%
  dplyr::group_by(Management, Block) %>%
  dplyr::summarise(
    ratio_prefer = mean(prefer[Warm == "Warming"]) / mean(prefer[Warm == "Ambient"]),
    .groups = "drop"
  ) %>%
  dplyr::group_by(Management) %>%
  dplyr::summarise(
    mean_value = mean(ratio_prefer),
    se_value = sd(ratio_prefer) / sqrt(dplyr::n()),
    .groups = "drop"
  ) %>%
  dplyr::mutate(
    treatment = factor(
      paste(Management, "ratio_prefer", sep = "_"),
      levels = c("Conventional_ratio_prefer", "Conservation_ratio_prefer")
    )
  ) %>%
  ggplot(aes(x = treatment, y = mean_value, shape = treatment)) +
  geom_errorbar(aes(ymin = mean_value - se_value, ymax = mean_value + se_value), width = 0.2,
  geom_point(size = 5, stroke = 1, fill = "white") +
  geom_hline(yintercept = 1, color = "grey50", linetype = "longdash", linewidth = 0.5) +
  scale_shape_manual(values = c(23, 21)) +
  scale_y_continuous(limits = c(0.95, 1.5)) +
  labs(y = "Warming effect", x = NULL) +
  my_theme +
  theme(legend.position = "none", axis.text.x = element_blank())
F.RAE_plot
```

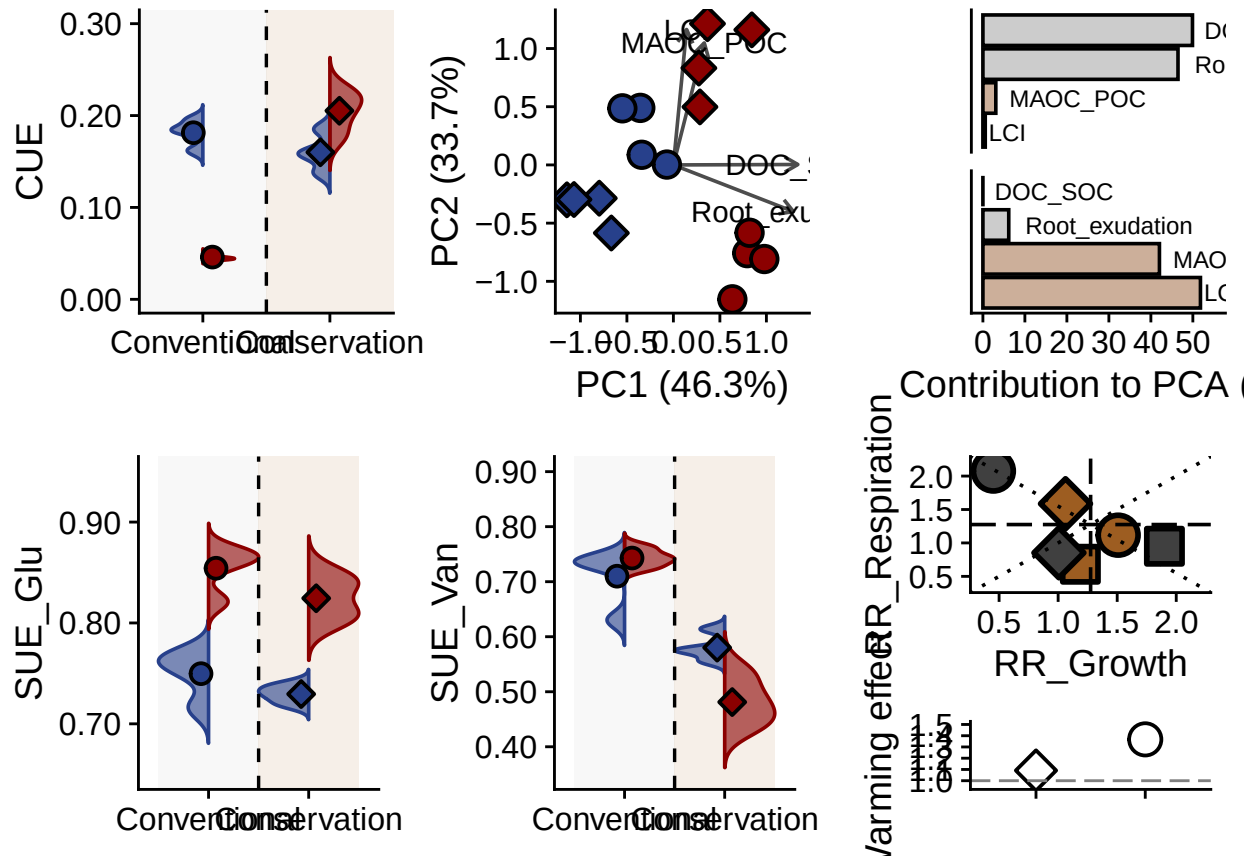



0.3.5 Figure assembly

```
Fig2.ABC_plot <- cowplot::plot_grid(
  A.CUE_plot,
  B.substrate.pca_plot,
  C.pca.contribution_plot,
  ncol = 3,
  align = "vh",
  axis = "tblr",
  rel_widths = c(2, 2, 2)
)

Fig2.DEF_plot <- cowplot::plot_grid(
  D.SUE.Glu_plot,
  E.SUE.Van_plot,
  (F.G.R_plot / F.RAE_plot) + patchwork::plot_layout(heights = c(2, 1)),
  ncol = 3,
  align = "vh",
  axis = "tblr",
  rel_widths = c(2, 2, 2)
)
```

```
Fig2.final_plot <- cowplot::plot_grid(
  Fig2.ABC_plot,
  Fig2.DEF_plot,
  nrow = 2,
  align = "hv",
  rel_heights = c(1, 1)
)
Fig2.final_plot
```



0.4 Fig3 Amplicon sequencing figures

```
df <- openxlsx::read.xlsx("data_for_RAE.xlsx", sheet = 2) %>%
  dplyr::mutate(
    Management = factor(Management, levels = c("Conventional", "Conservation"))
  ); head(df)
```

##	Treat	Management	Warm	Block	B_Richness	B_Shannon	F_Richness	F_Shannon
## 1	TN1	Conventional	Ambient	B1	1576	9.8057	662	6.7147
## 2	TN2	Conventional	Ambient	B2	1748	9.9511	617	7.0963
## 3	TN3	Conventional	Ambient	B3	1766	9.9560	598	6.6826

	Treatment	Phylum	rrn_copy	Acidobacteriota	Actinomycetota	Bacillota
## 4	TN4 Conventional Ambient	B4	1846	9.9553	620	6.8605
## 5	TW1 Conventional Warming	B1	1966	10.0255	574	6.6722
## 6	TW2 Conventional Warming	B2	1934	10.0575	621	6.8160
##	B_r_K	F_r_K	rrn_copy	Acidobacteriota	Actinomycetota	Bacillota
## 1	1.238791	23.65689	2.348447	0.1365677	0.2479124	0.08680947
## 2	1.095523	17.17802	2.340382	0.1344849	0.2304473	0.08434449
## 3	1.080784	35.28540	2.335465	0.1365677	0.2486385	0.07188581
## 4	1.209176	25.37000	2.258449	0.1229243	0.2923968	0.05717234
## 5	1.281785	35.61974	2.322594	0.1054975	0.3074542	0.05929337
## 6	1.164499	21.76425	2.245732	0.1250836	0.2802629	0.06034433
##	Bacteroidota	Chloroflexota	Gemmatimonadota	Myxococcota	B_Others	
## 1	0.06699406	0.07771387	0.01545870	0.03265626	0.06766285	
## 2	0.06365009	0.08147823	0.01698737	0.03019128	0.06156727	
## 3	0.05814687	0.07263104	0.01920394	0.03051612	0.05340798	
## 4	0.04748438	0.07607055	0.02063707	0.02850974	0.05413410	
## 5	0.05224237	0.07257371	0.01952879	0.02801292	0.04943344	
## 6	0.05566277	0.07352913	0.01996828	0.03114670	0.05017866	
##	Planctomycetota	Pseudomonadota	Ascomycota	Basidiomycota	Blastocladiomycota	
## 1	0.06212141	0.2061032	0.6355314	0.02686454	0.0002627596	
## 2	0.07746546	0.2193836	0.5831687	0.03394854	0.0003468427	
## 3	0.08329352	0.2257084	0.6653283	0.01885563	0.0002207181	
## 4	0.06840808	0.2322626	0.6280095	0.02655624	0.0002767735	
## 5	0.07798139	0.2279823	0.6409337	0.01799378	0.0003363323	
## 6	0.07440812	0.2294155	0.5657004	0.02599218	0.0005360296	
##	Chytridiomycota	Glomeromycota	Kickxellomycota	Mortierellomycota	Mucoromycota	
## 1	0.03111074	0.0006096023	1.576558e-04	0.05279366	0.03719625	
## 2	0.03395905	0.0006411334	3.153115e-05	0.04576221	0.02834651	
## 3	0.02587657	0.0001261246	8.408307e-05	0.03646052	0.02122047	
## 4	0.03031545	0.0004589534	9.109000e-05	0.04500547	0.02892107	
## 5	0.03007021	0.0001681661	1.471454e-04	0.03403262	0.10412638	
## 6	0.03620827	0.0005360296	5.255192e-05	0.04047549	0.13652989	
##	Olpidiomycota	F_Others				
## 1	5.255192e-05	0.2154208				
## 2	7.357269e-05	0.2737219				
## 3	4.204154e-05	0.2317855				
## 4	5.605538e-05	0.2403094				
## 5	0.000000e+00	0.1721916				
## 6	3.153115e-05	0.1939376				

```

phylum.sample_order <- c(
  paste0("TN", 1:4),
  paste0("TW", 1:4),
  paste0("CN", 1:4),
  paste0("CW", 1:4)
)

df$Treat <- factor(df$Treat, levels = phylum.sample_order)

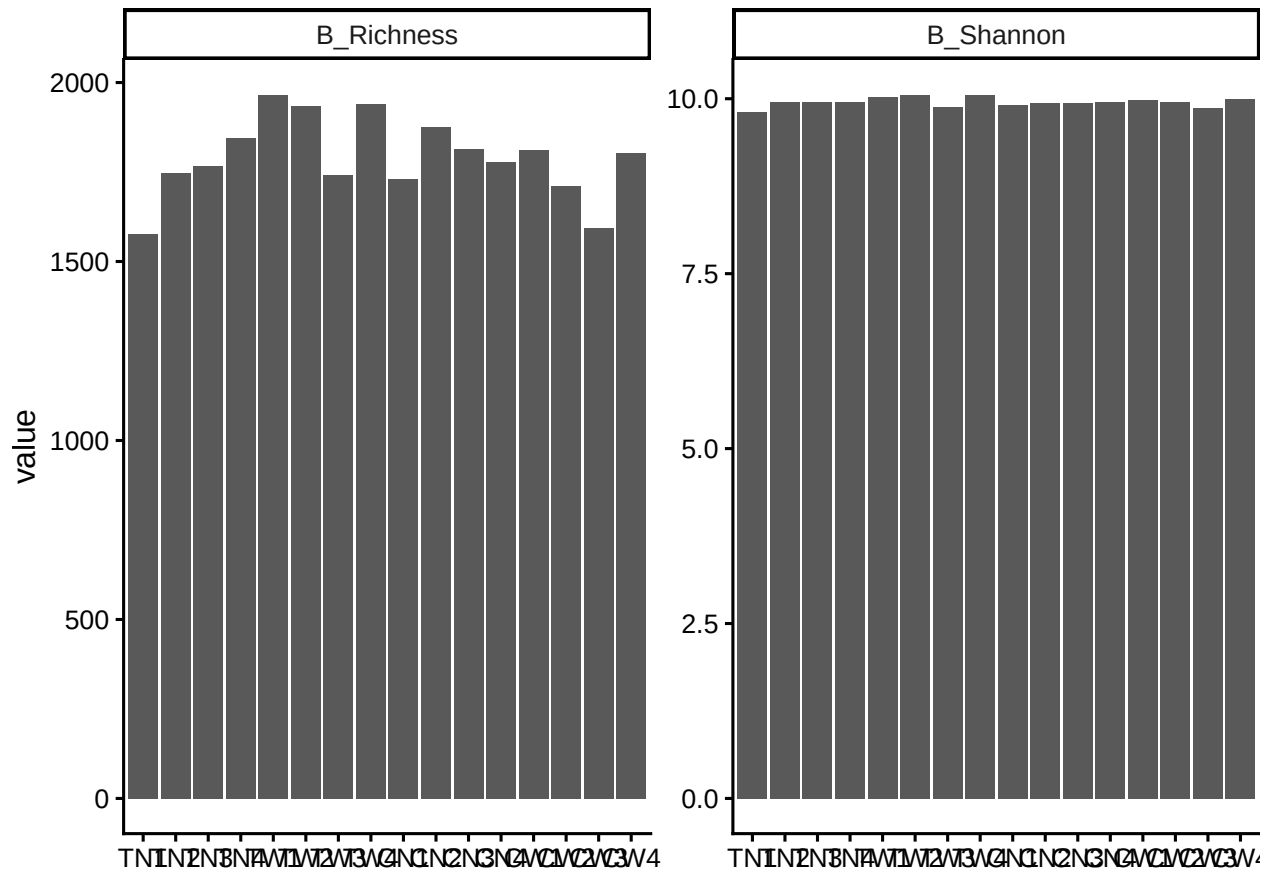
```

0.4.1 A and B. Alpha diversity

```
bacteria.alpha_df <- df %>%  
  dplyr::select(Treat, B_Richness, B_Shannon) %>%  
  tidyr::pivot_longer(-Treat)  
head(bacteria.alpha_df)
```

```
## # A tibble: 6 x 3  
##   Treat name      value  
##   <fct> <chr>      <dbl>  
## 1 TN1    B_Richness 1576  
## 2 TN1    B_Shannon   9.81  
## 3 TN2    B_Richness 1748  
## 4 TN2    B_Shannon   9.95  
## 5 TN3    B_Richness 1766  
## 6 TN3    B_Shannon   9.96
```

```
bacteria.alpha_plot <- ggplot(bacteria.alpha_df, aes(x = Treat, y = value)) +  
  geom_bar(stat = "identity") +  
  facet_wrap(~name, scales = "free") +  
  labs(x = NULL) +  
  ggh4x::facetted_pos_scales(  
    y = list(  
      name == "Richness" ~ scale_y_continuous(limits = c(1200, 2000), breaks = c(1200, 2000)),  
      name == "Shannon" ~ scale_y_continuous(limits = c(8, 11), breaks = c(8, 11))  
    )  
  ) +  
  theme_classic()  
bacteria.alpha_plot
```



```
fungi.alpha_df <- df %>%
  dplyr::select(Treat, F_Richness, F_Shannon) %>%
  tidyr::pivot_longer(-Treat)
head(fungi.alpha_df)
```

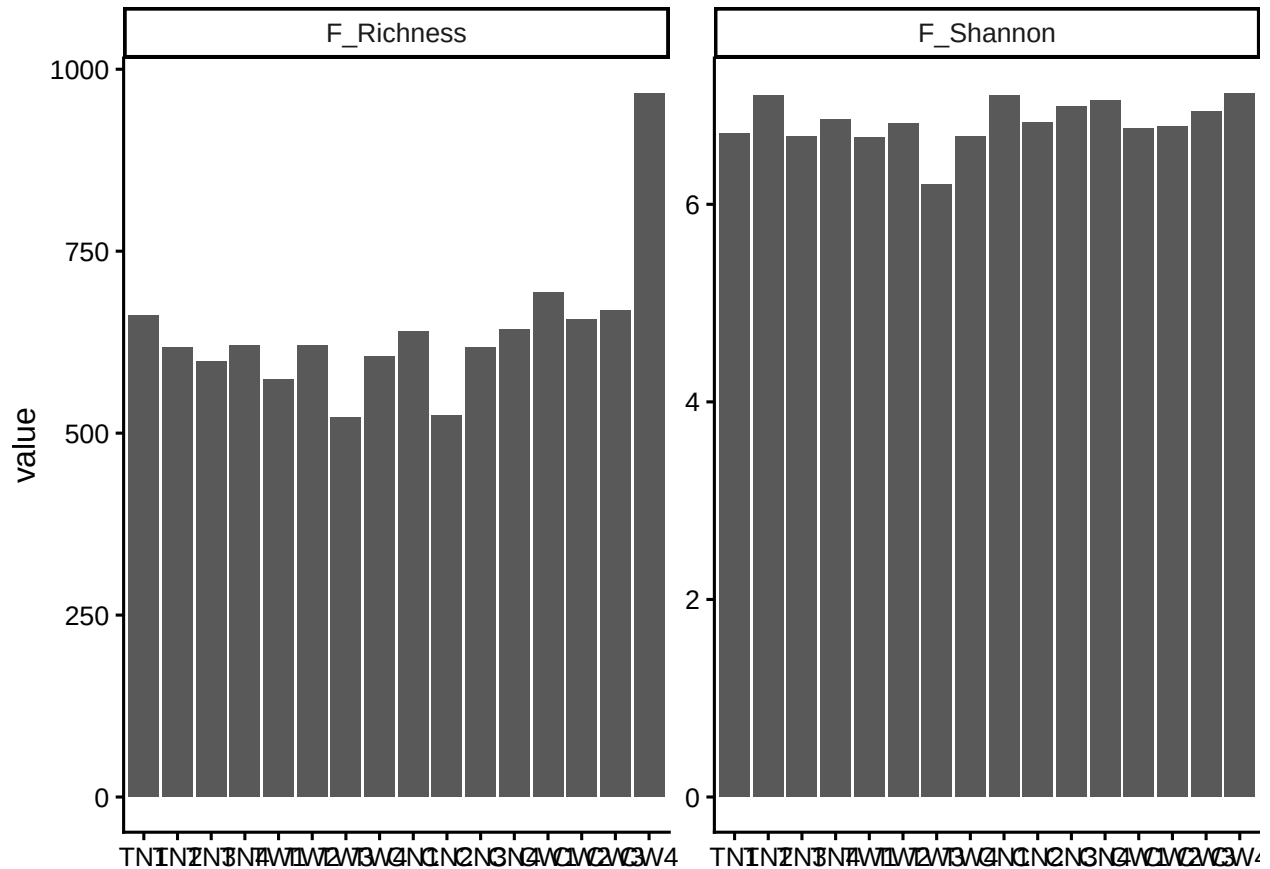
```
## # A tibble: 6 x 3
##   Treat name      value
##   <fct> <chr>      <dbl>
## 1 TN1     F_Richness 662
## 2 TN1     F_Shannon  6.71
## 3 TN2     F_Richness 617
## 4 TN2     F_Shannon  7.10
## 5 TN3     F_Richness 598
## 6 TN3     F_Shannon  6.68
```

```
fungi.alpha_plot <- ggplot(fungi.alpha_df, aes(x = Treat, y = value)) +
  geom_bar(stat = "identity") +
  facet_wrap(~name, scales = "free") +
  labs(x = NULL) +
  ggh4x::facetted_pos_scales(
    y = list(
      name == "Richness" ~ scale_y_continuous(limits = c(300, 1000), breaks = c(300, 1000)),
```

```

    name == "Shannon" ~ scale_y_continuous(limits = c(4, 8), breaks = c(4, 8))
  )
) +
  theme_classic()
fungi.alpha_plot

```



0.4.2 A and B. Bacterial species composition at phylum level

```

bac.phylum.comp_df <- df %>%
  dplyr::select(Treat, Acidobacteriota:Pseudomonadota) %>%
  tidyr::pivot_longer(
    cols = Acidobacteriota:Pseudomonadota,
    names_to = "ID",
    values_to = "Abundance"
  ) %>%
  tidyr::pivot_wider(
    names_from = Treat,
    values_from = Abundance,
    names_sort = FALSE
  )
head(bac.phylum.comp_df)

```

```
## # A tibble: 6 x 17
##   ID      TN1      TN2      TN3      TN4      TW1      TW2      TW3      TW4      CN1      CN2
##   <chr>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 Acidoba~ 0.137    0.134    0.137    0.123    0.105    0.125    0.114    0.121    0.117    0.108
## 2 Actinom~ 0.248    0.230    0.249    0.292    0.307    0.280    0.264    0.256    0.295    0.298
## 3 Bacillo~ 0.0868   0.0843   0.0719   0.0572   0.0593   0.0603   0.0627   0.0694   0.0793   0.0776
## 4 Bactero~ 0.0670   0.0637   0.0581   0.0475   0.0522   0.0557   0.0737   0.0612   0.0629   0.0524
## 5 Chlorof~ 0.0777   0.0815   0.0726   0.0761   0.0726   0.0735   0.0717   0.0815   0.0599   0.0577
## 6 Gemmati~ 0.0155   0.0170   0.0192   0.0206   0.0195   0.0200   0.0174   0.0201   0.0173   0.0123
## # i 6 more variables: CN3 <dbl>, CN4 <dbl>, CW1 <dbl>, CW2 <dbl>, CW3 <dbl>,
## #   CW4 <dbl>
```

```
bac.phylum.cols <- c(
  "#294b41", "#D9B5CF", "#3A9B3F", "#C7252D", "#28A9C9",
  "#F28E2B", "#7B5AA6", "#4F6FAE", "#8AA0A8", "#4D4D4D"
)

bac.phylum.top9_taxa <- c(
  "Actinomycetota", "Pseudomonadota", "Acidobacteriota",
  "Planctomycetota", "Bacillota", "Chloroflexota",
  "Bacteroidota", "Myxococcota", "Gemmatimonadota"
)

names(bac.phylum.cols) <- c(bac.phylum.top9_taxa, "B_Others")

plot.bac.phylum_comp <- function(sample.pattern, group.name) {
  bac.phylum.sample_order <- phylum.sample_order[
    stringr::str_detect(phylum.sample_order, sample.pattern)
  ]

  bac.phylum.sub_df <- bac.phylum.comp_df %>%
    dplyr::select(ID, dplyr::all_of(bac.phylum.sample_order))

  bac.phylum.long_df <- bac.phylum.sub_df %>%
    tidyr::pivot_longer(-ID, names_to = "Sample", values_to = "Abundance") %>%
    dplyr::group_by(Sample) %>%
    dplyr::mutate(Percent = Abundance / sum(Abundance, na.rm = TRUE) * 100) %>%
    dplyr::ungroup() %>%
    dplyr::mutate(
      Sample = factor(Sample, levels = bac.phylum.sample_order),
      ID = factor(ID, levels = rev(c(bac.phylum.top9_taxa, "B_Others")))
    )

  ggplot(bac.phylum.long_df, aes(Sample, Percent, fill = ID)) +
    geom_col(width = 0.85, color = "white", linewidth = 0.15) +
    coord_flip(ylim = c(0, 100), clip = "off") +
    scale_fill_manual(values = bac.phylum.cols, breaks = names(bac.phylum.cols)) +
```

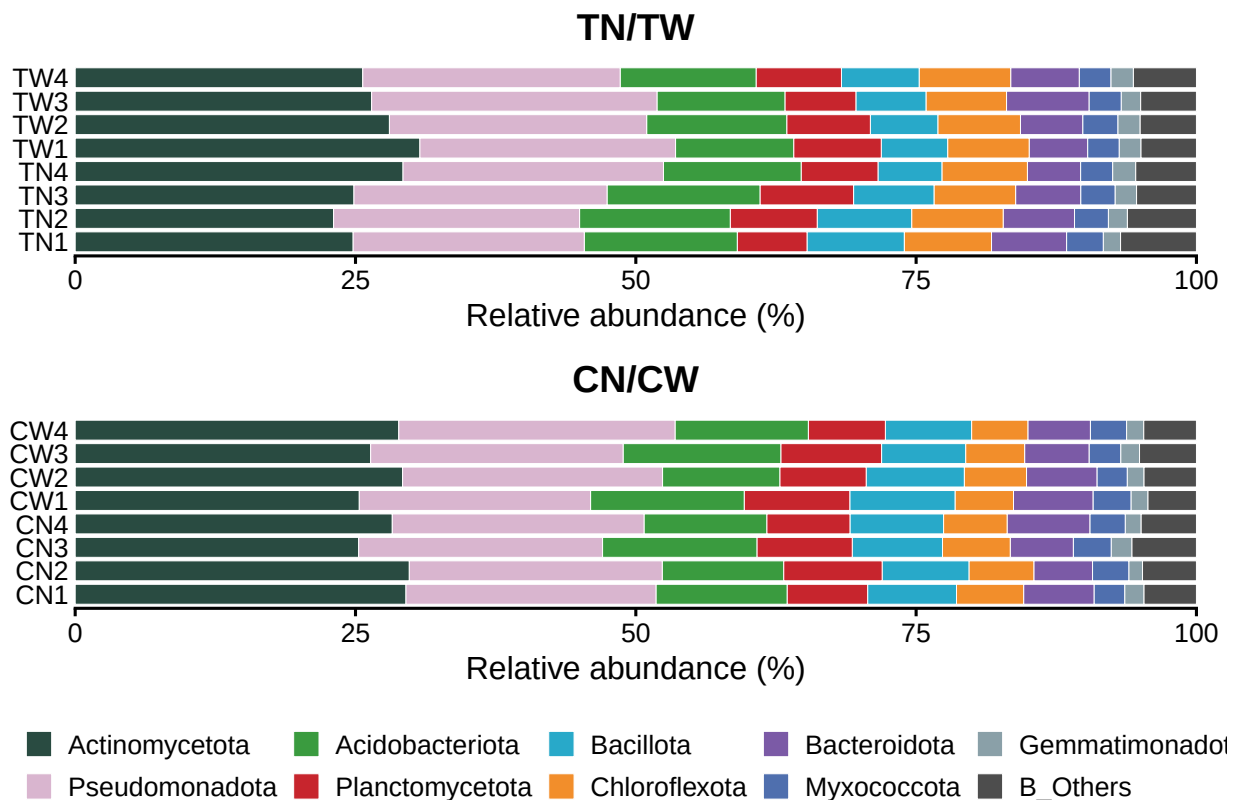
```

scale_y_continuous(expand = c(0, 0), breaks = seq(0, 100, 25)) +
labs(x = NULL, y = "Relative abundance (%)", fill = NULL, title = group.name) +
theme_classic(base_size = 12) +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold"),
  axis.text.y = element_text(color = "black"),
  axis.ticks.y = element_blank(),
  axis.line.y = element_blank(),
  legend.position = "bottom",
  legend.key.size = unit(0.35, "cm"),
  legend.text = element_text(size = 10)
)
}

bac.phylum.CNCW_plot <- plot.bac.phylum_comp("^ (CN|CW)", "CN/CW")
bac.phylum.TNTW_plot <- plot.bac.phylum_comp("^ (TN|TW)", "TN/TW")

bac.phylum.combined_plot <- (bac.phylum.TNTW_plot / bac.phylum.CNCW_plot) +
  patchwork::plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
bac.phylum.combined_plot

```



0.4.3 A and B. Fungal species composition at phylum level

```
fungi.phylum.comp_df <- df %>%
  dplyr::select(Treat, Ascomycota:F_Others) %>%
  tidyr::pivot_longer(
    cols = Ascomycota:F_Others,
    names_to = "ID",
    values_to = "Abundance"
  ) %>%
  tidyr::pivot_wider(
    names_from = Treat,
    values_from = Abundance,
    names_sort = FALSE
  )
head(fungi.phylum.comp_df)
```

```
## # A tibble: 6 x 17
##   ID      TN1      TN2      TN3      TN4      TW1      TW2      TW3      TW4      CN1
##   <chr>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1 Ascom~ 6.36e-1 5.83e-1 6.65e-1 6.28e-1 6.41e-1 5.66e-1 6.07e-1 6.15e-1 6.85e-1
## 2 Basid~ 2.69e-2 3.39e-2 1.89e-2 2.66e-2 1.80e-2 2.60e-2 2.07e-2 1.81e-2 3.89e-2
## 3 Blast~ 2.63e-4 3.47e-4 2.21e-4 2.77e-4 3.36e-4 5.36e-4 4.59e-4 5.04e-4 3.05e-4
## 4 Chytr~ 3.11e-2 3.40e-2 2.59e-2 3.03e-2 3.01e-2 3.62e-2 3.65e-2 4.31e-2 4.24e-2
## 5 Glome~ 6.10e-4 6.41e-4 1.26e-4 4.59e-4 1.68e-4 5.36e-4 3.05e-4 2.10e-4 2.45e-3
## 6 Kickx~ 1.58e-4 3.15e-5 8.41e-5 9.11e-5 1.47e-4 5.26e-5 7.36e-5 2.10e-5 1.05e-4
## # i 7 more variables: CN2 <dbl>, CN3 <dbl>, CN4 <dbl>, CW1 <dbl>, CW2 <dbl>,
## #   CW3 <dbl>, CW4 <dbl>
```

```
fungi.phylum.cols <- c(
  "#2F80C9", "#D98627", "#8C77B5", "#4AB0A7", "#79B9E3",
  "#9E2428", "#51338D", "#9A6334", "#1D6264", "#D9D9D9"
)

fungi.phylum.top9_taxa <- c(
  "Ascomycota", "Basidiomycota", "Blastocladiomycota",
  "Chytridiomycota", "Glomeromycota", "Kickxellomycota",
  "Mortierellomycota", "Mucoromycota", "Olpidiomycota"
)

names(fungi.phylum.cols) <- c(fungi.phylum.top9_taxa, "F_Others")

plot.fungi.phylum_comp <- function(sample.pattern, group.name) {
  fungi.phylum.sample_order <- phylum.sample_order[
    stringr::str_detect(phylum.sample_order, sample.pattern)
  ]
}
```

```

fungi.phylum.sub_df <- fungi.phylum.comp_df %>%
  dplyr::select(ID, dplyr::all_of(fungi.phylum.sample_order))

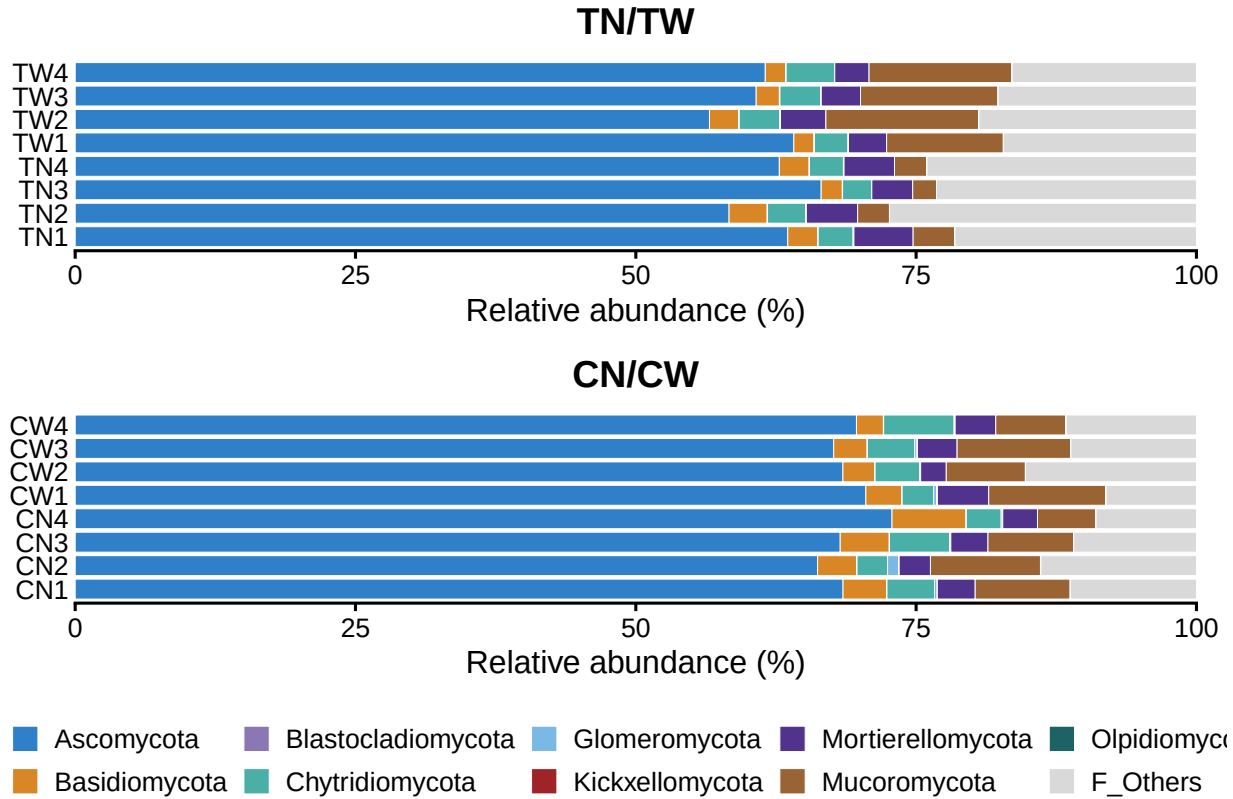
fungi.phylum.long_df <- fungi.phylum.sub_df %>%
  tidyr::pivot_longer(-ID, names_to = "Sample", values_to = "Abundance") %>%
  dplyr::group_by(Sample) %>%
  dplyr::mutate(Percent = Abundance / sum(Abundance, na.rm = TRUE) * 100) %>%
  dplyr::ungroup() %>%
  dplyr::mutate(
    Sample = factor(Sample, levels = fungi.phylum.sample_order),
    ID = factor(ID, levels = rev(c(fungi.phylum.top9_taxa, "F_Others")))
  )

ggplot(fungi.phylum.long_df, aes(Sample, Percent, fill = ID)) +
  geom_col(width = 0.85, color = "white", linewidth = 0.15) +
  coord_flip(ylim = c(0, 100), clip = "off") +
  scale_fill_manual(values = fungi.phylum.cols, breaks = names(fungi.phylum.cols)) +
  scale_y_continuous(expand = c(0, 0), breaks = seq(0, 100, 25)) +
  labs(x = NULL, y = "Relative abundance (%)", fill = NULL, title = group.name) +
  theme_classic(base_size = 12) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.text.y = element_text(color = "black"),
    axis.ticks.y = element_blank(),
    axis.line.y = element_blank(),
    legend.position = "bottom",
    legend.key.size = unit(0.35, "cm"),
    legend.text = element_text(size = 10)
  )
}

fungi.phylum.CNCW_plot <- plot.fungi.phylum_comp("^ (CN|CW)", "CN/CW")
fungi.phylum.TNTW_plot <- plot.fungi.phylum_comp("^ (TN|TW)", "TN/TW")

fungi.phylum.combined_plot <- (fungi.phylum.TNTW_plot / fungi.phylum.CNCW_plot) +
  patchwork::plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
fungi.phylum.combined_plot

```



0.4.4 C-E. Life-history strategies based on amplicon sequencing

```
plot.box_trait <- function(data_df, y.var, y.lim = NULL, show.x.text = FALSE) {
  y.var <- rlang::enquo(y.var)

  trait_plot <- data_df %>%
    ggplot(aes(Management, !!y.var, fill = Warm, shape = Management)) +
    geom_rect(
      data = background_data,
      aes(xmin = xmin, xmax = xmax, ymin = -Inf, ymax = Inf),
      fill = background_data$fill_color,
      alpha = 0.1,
      inherit.aes = FALSE
    ) +
    geom_boxplot(
      alpha = 0.6,
      width = 0.5,
      outlier.shape = NA,
      coef = 2,
      position = position_dodge(width = 0.6)
    ) +
}
```

```

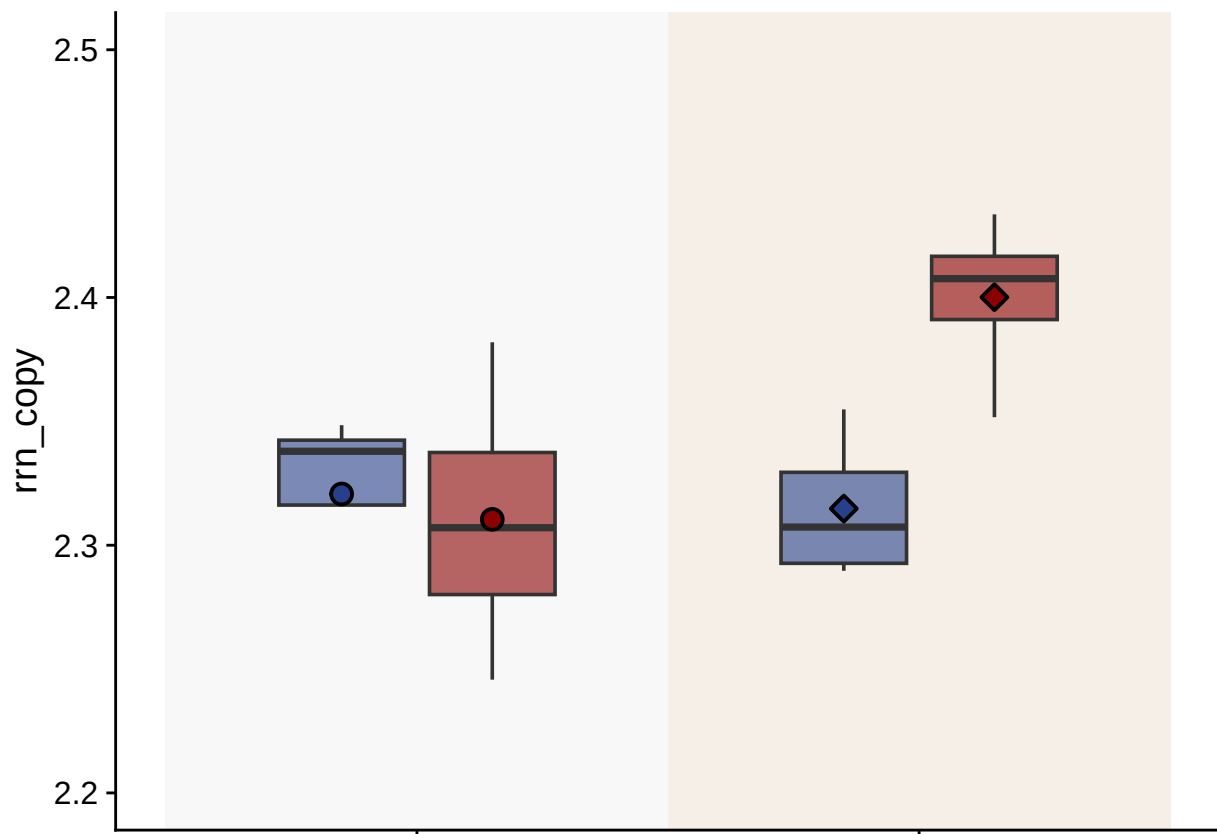
stat_summary(
  fun = "mean",
  geom = "point",
  position = position_dodge(width = 0.6),
  show.legend = FALSE,
  size = 3,
  color = "black",
  stroke = 1
) +
scale_fill_manual(values = cols1) +
scale_shape_manual(values = c(21, 23)) +
my_theme +
theme(axis.title.x = element_blank(), legend.position = "none")

if (!is.null(y.lim)) trait_plot <- trait_plot + coord_cartesian(ylim = y.lim)
if (!show.x.text) trait_plot <- trait_plot + theme(axis.text.x = element_blank())

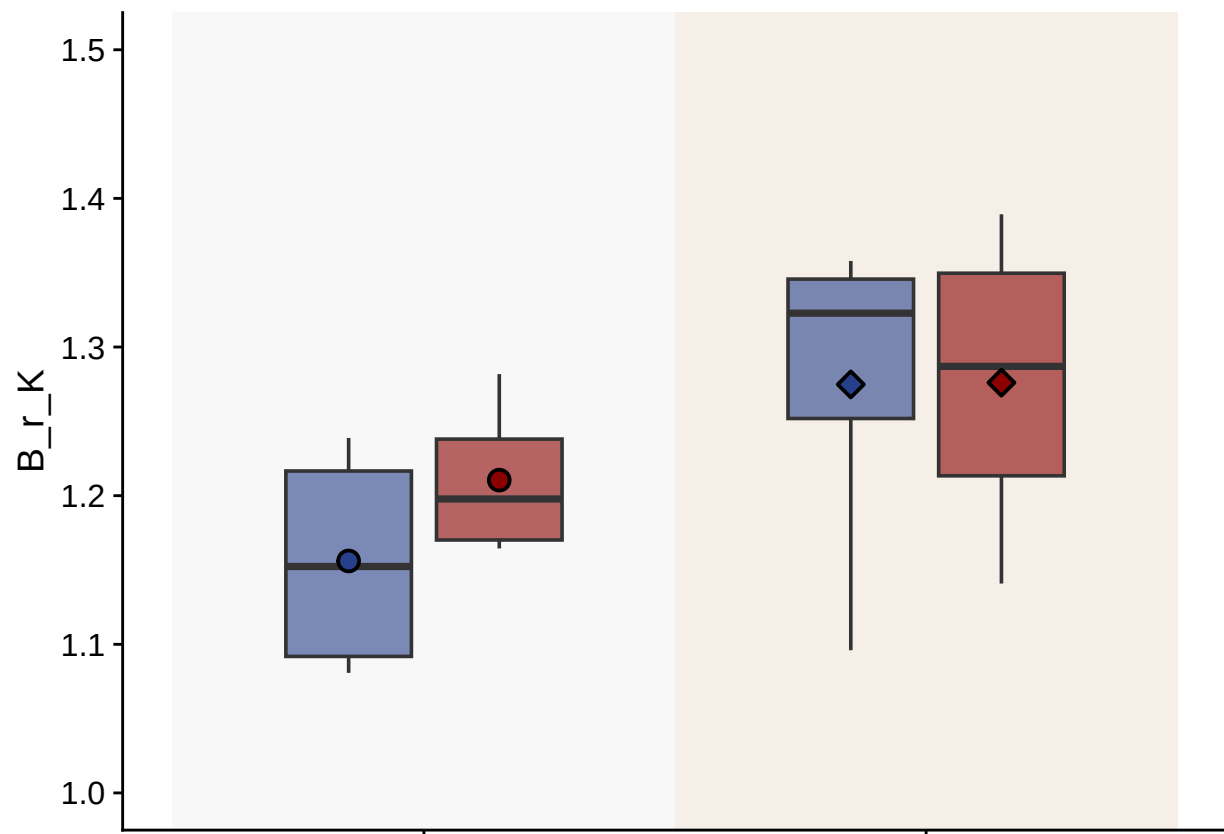
trait_plot
}

rrn.copy_plot <- plot.box_trait(df, rrn_copy, show.x.text = FALSE) +
  scale_y_continuous(limits = c(2.2, 2.5))
rrn.copy_plot

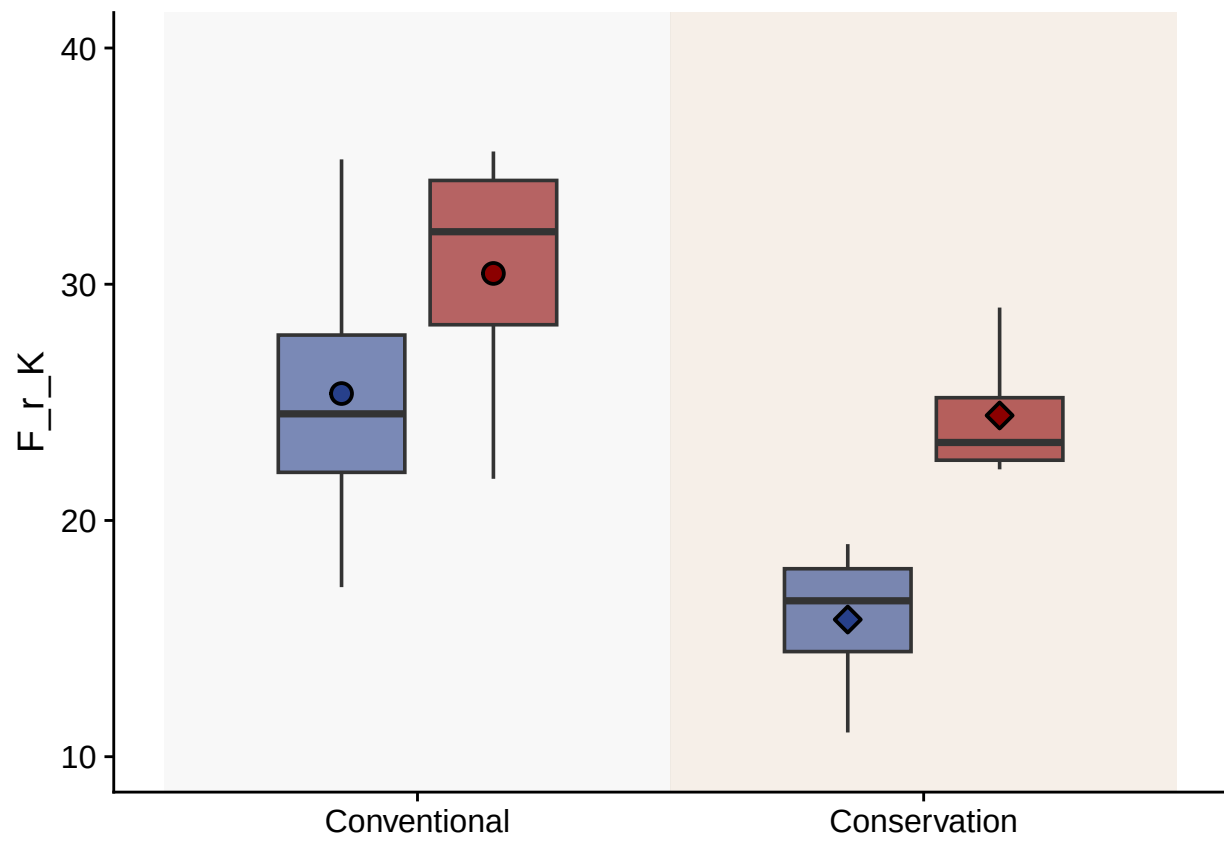
```



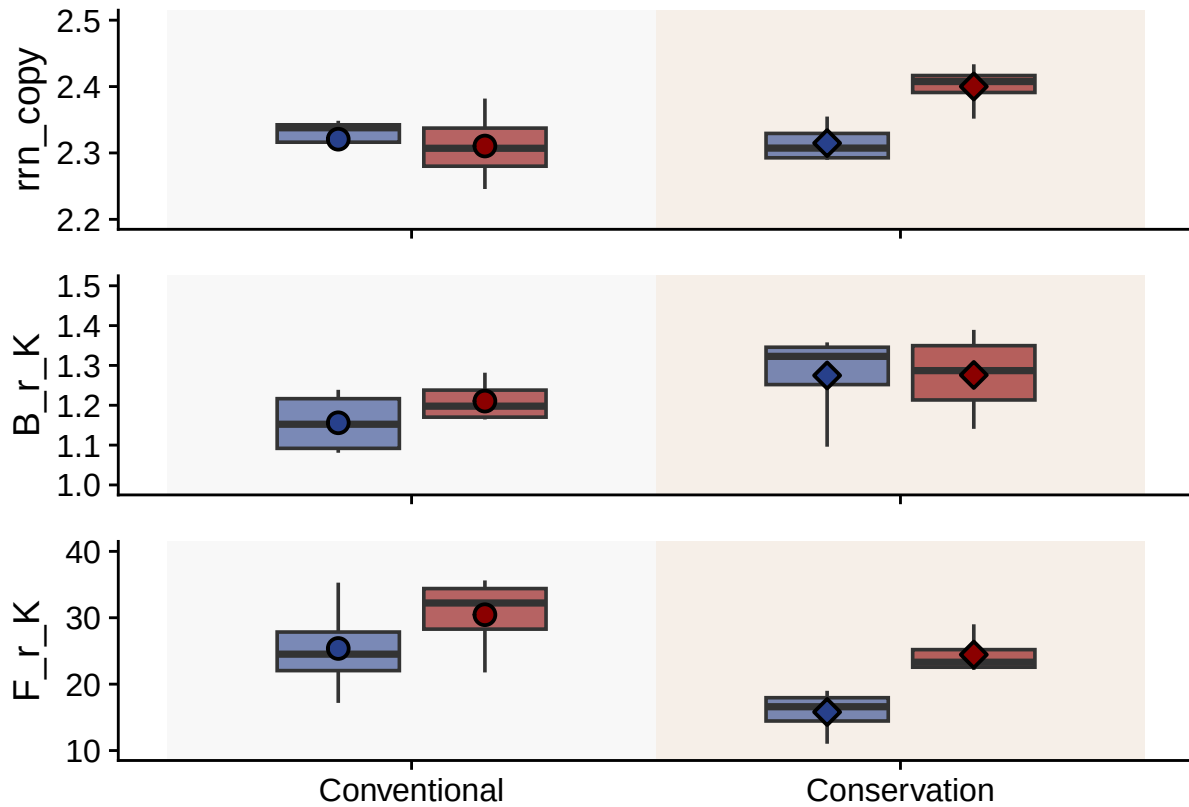
```
B.rK_plot <- plot.box_trait(df, B_r_K, show.x.text = FALSE) +  
  coord_cartesian(ylim = c(1.0, 1.5))  
B.rK_plot
```



```
F.rK_plot <- plot.box_trait(df, F_r_K, show.x.text = TRUE) +  
  scale_y_continuous(limits = c(10, 40))  
F.rK_plot
```



```
life.history_plot <- rrn.copy_plot / B.rK_plot / F.rK_plot  
life.history_plot
```



0.5 Fig4 Metagenome analysis

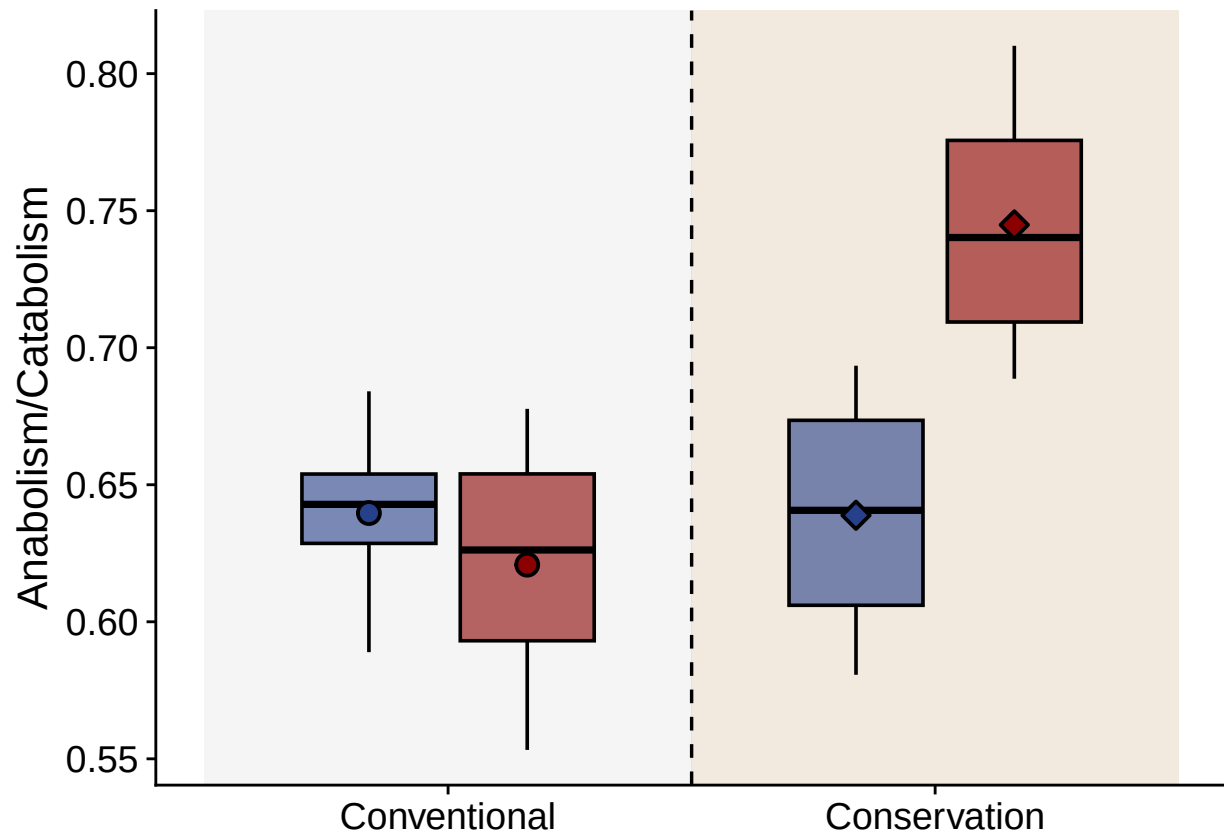
0.5.1 B. functional profile

```
df <- read.xlsx("data_for_RAE.xlsx", sheet = 3) |>
  mutate(Management = factor(Management, levels = c('Conventional', 'Conservation')),
         Warm = factor(Warm), AGS_Mb = AGS / 1e6); head(df)
```

##	Treat	Management	Warm	Block	A_C_ratio	Gene_r_K	AGS	CUBHE
## 1	TN1	Conventional	Ambient	B1	0.5889209	6.179635	5547158	0.7534295
## 2	TN2	Conventional	Ambient	B2	0.6840892	8.266193	5586851	0.7548454
## 3	TN3	Conventional	Ambient	B3	0.6418357	6.224945	5593732	0.7532773
## 4	TN4	Conventional	Ambient	B4	0.6438399	7.544028	5669604	0.7709763
## 5	TW1	Conventional	Warming	B1	0.6063588	7.805807	5702845	0.5985978
## 6	TW2	Conventional	Warming	B2	0.6460347	6.714818	5469188	0.7756629
##	ACN	d	AGS_Mb					
## 1	1.388147	10.216358	5.547158					
## 2	1.408330	8.087823	5.586851					
## 3	1.399719	8.478932	5.593732					
## 4	1.422273	10.524900	5.669604					
## 5	1.447570	8.626968	5.702845					


```
## 6 1.442050 10.747814 5.469188
```

```
A.C.ratio_plot <- ggplot(df, aes(x = Management, y = A_C_ratio, fill = Warm, shape = Management)) +  
  geom_rect(  
    data = background_data,  
    aes(xmin = xmin, xmax = xmax, ymin = -Inf, ymax = Inf),  
    fill = background_data$fill_color,  
    alpha = 0.15,  
    inherit.aes = FALSE  
  ) +  
  geom_boxplot(  
    alpha = .6,  
    width = .55,  
    outlier.shape = NA,  
    coef = 2,  
    position = position_dodge(.65),  
    color = "black"  
  ) +  
  stat_summary(  
    fun = "mean",  
    geom = "point",  
    position = position_dodge(.65),  
    show.legend = FALSE,  
    size = 3.2,  
    color = "black",  
    stroke = 1  
  ) +  
  geom_vline(xintercept = 1.5, linetype = "dashed", color = "black", linewidth = 0.6) +  
  scale_fill_manual(values = cols1) +  
  scale_shape_manual(values = c(21, 23)) +  
  labs(x = NULL, y = "Anabolism/Catabolism", fill = NULL) +  
  theme_cowplot() +  
  theme(  
    axis.text = element_text(size = 14),  
    axis.title.y = element_text(size = 16),  
    legend.position = "none"  
  )  
A.C.ratio_plot
```

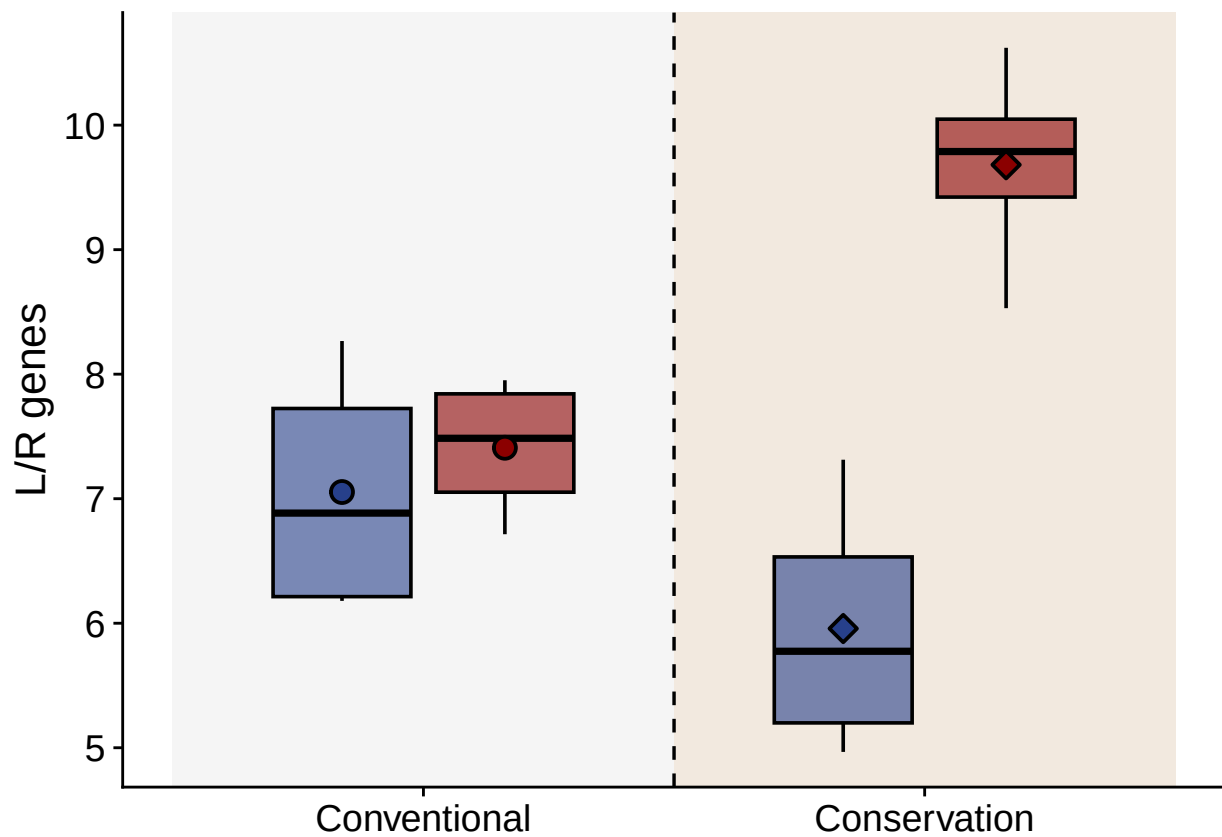


```
Gene.rK_plot <- ggplot(df, aes(x = Management, y = Gene_r_K, fill = Warm, shape = Management))
  geom_rect(
    data = background_data,
    aes(xmin = xmin, xmax = xmax, ymin = -Inf, ymax = Inf),
    fill = background_data$fill_color,
    alpha = 0.15,
    inherit.aes = FALSE
  ) +
  geom_boxplot(
    alpha = .6,
    width = .55,
    outlier.shape = NA,
    coef = 2,
    position = position_dodge(.65),
    color = "black"
  ) +
  stat_summary(
    fun = "mean",
    geom = "point",
    position = position_dodge(.65),
    show.legend = FALSE,
    size = 3.2,
  )
```

```

    color = "black",
    stroke = 1
) +
geom_vline(xintercept = 1.5, linetype = "dashed", color = "black", linewidth = 0.6) +
scale_fill_manual(values = cols1) +
scale_shape_manual(values = c(21, 23)) +
labs(x = NULL, y = "L/R genes", fill = NULL) +
theme_cowplot() +
theme(
  axis.text = element_text(size = 14),
  axis.title.y = element_text(size = 16),
  legend.position = "none"
)
Gene.rK_plot

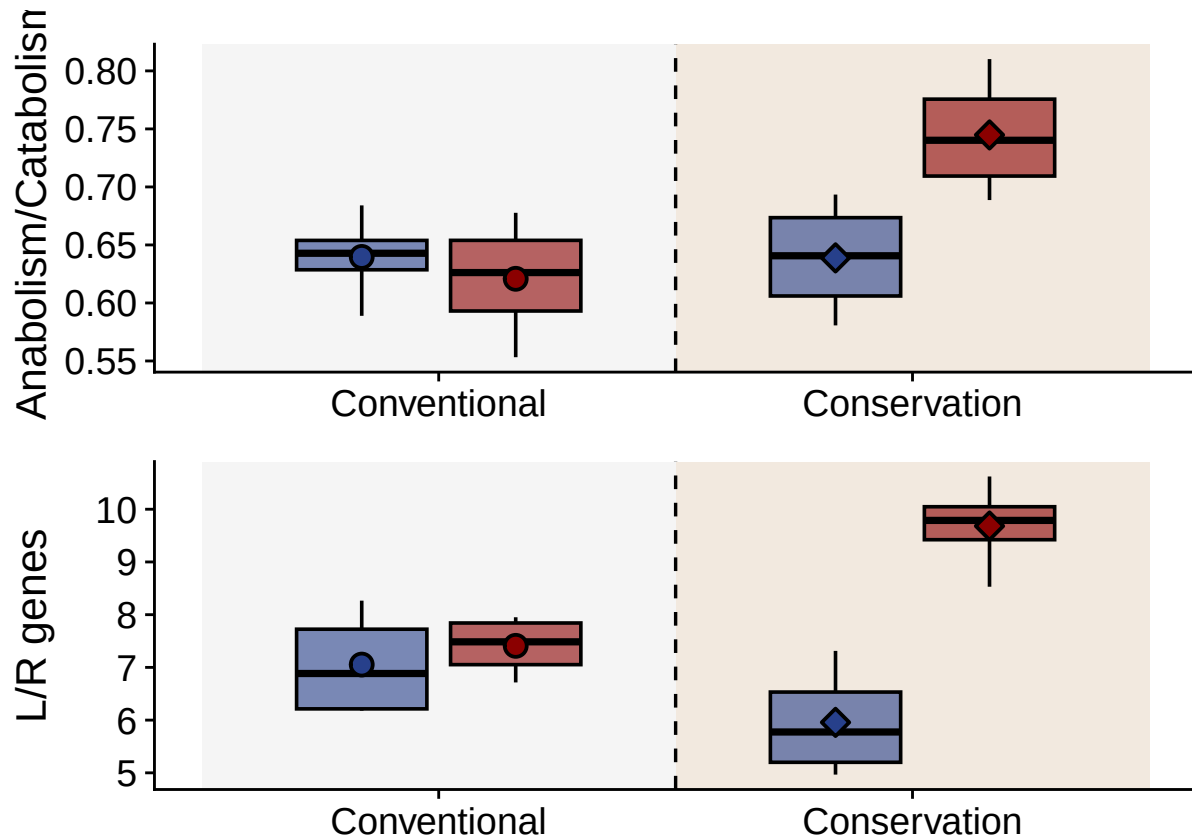
```



```

functional.profile_plot <- A.C.ratio_plot / Gene.rK_plot
functional.profile_plot

```



0.5.2 C. microbial life-history traits at contigs level

```
vars <- c("AGS_Mb" = "Average genome size (Mbp)",
          "ACN" = "Average 16S copy number",
          "CUBHE" = "CUBHE",
          "d" = "Minimal doubling time (h-1)")

plots <- list()

for (v in names(vars)) {
  df_v <- df

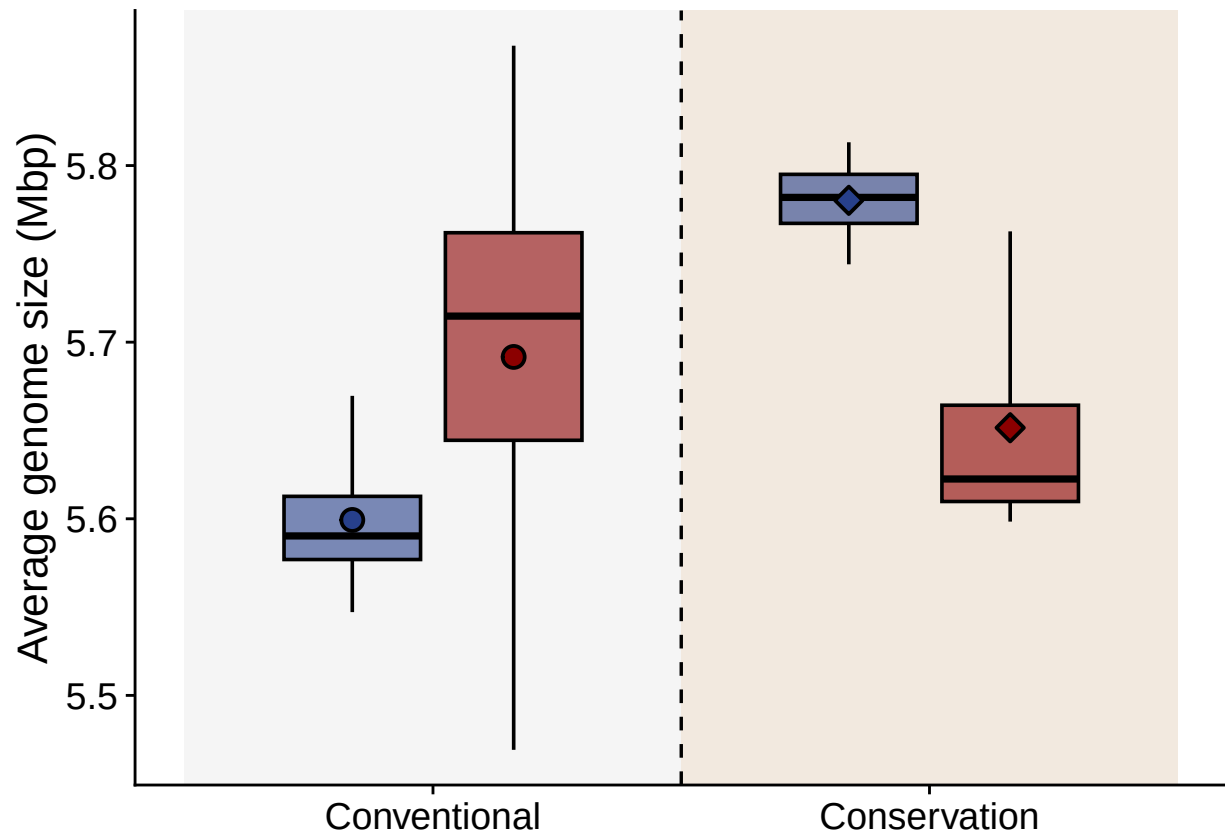
  plots[[v]] <- ggplot(df_v, aes(x = Management, y = .data[[v]], fill = Warm, shape = Management)) +
    geom_rect(data = background_data, aes(xmin = xmin, xmax = xmax, ymin = -Inf, ymax = Inf), fill = "white",
              color = "black", linewidth = 0.5) +
    geom_boxplot(alpha = .6, width = .55, outlier.shape = NA, coef = 2, position = position_dodge(.65)) +
    stat_summary(fun = "mean", geom = "point", position = position_dodge(.65), show.legend = FALSE) +
    geom_vline(xintercept = 1.5, linetype = "dashed", color = "black", linewidth = 0.6) +
    scale_fill_manual(values = cols1) +
    scale_shape_manual(values = c(21, 23)) +
    labs(x = NULL, y = vars[v], fill = NULL) +
    theme_cowplot() +
```

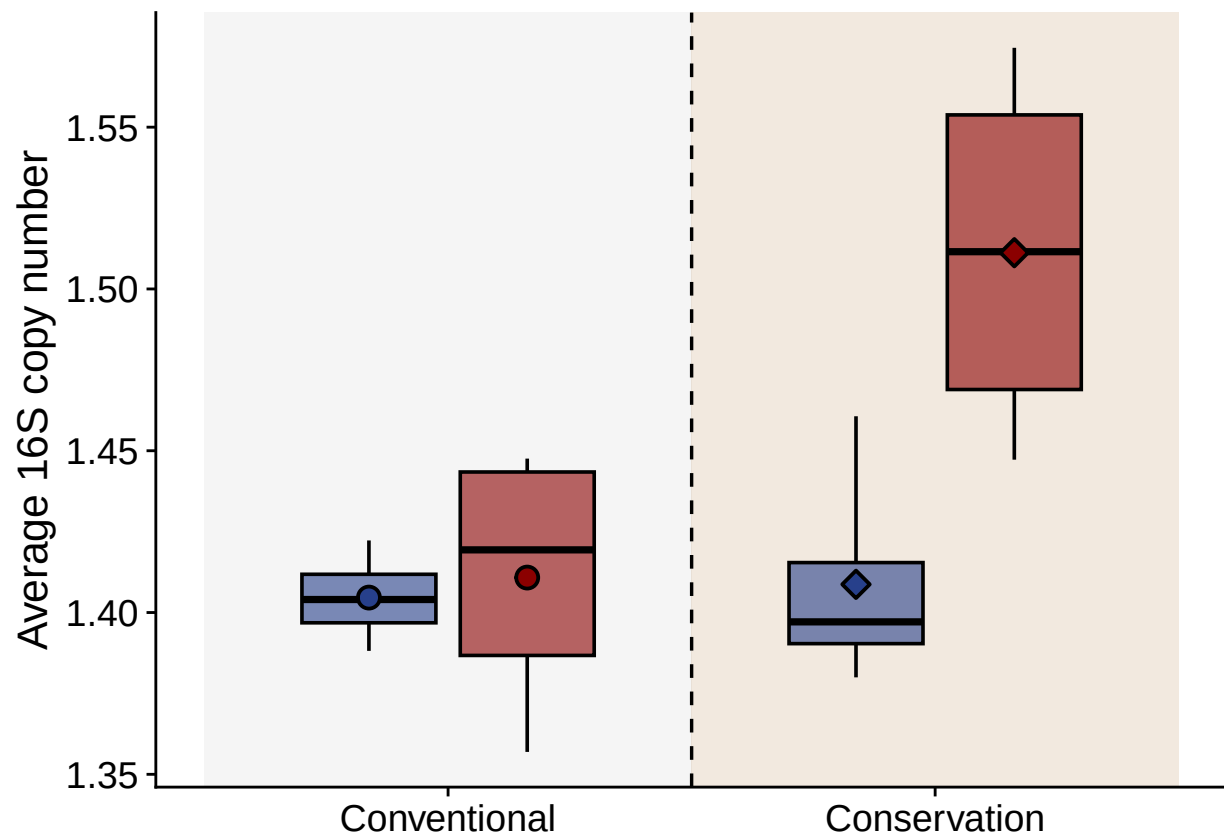
```

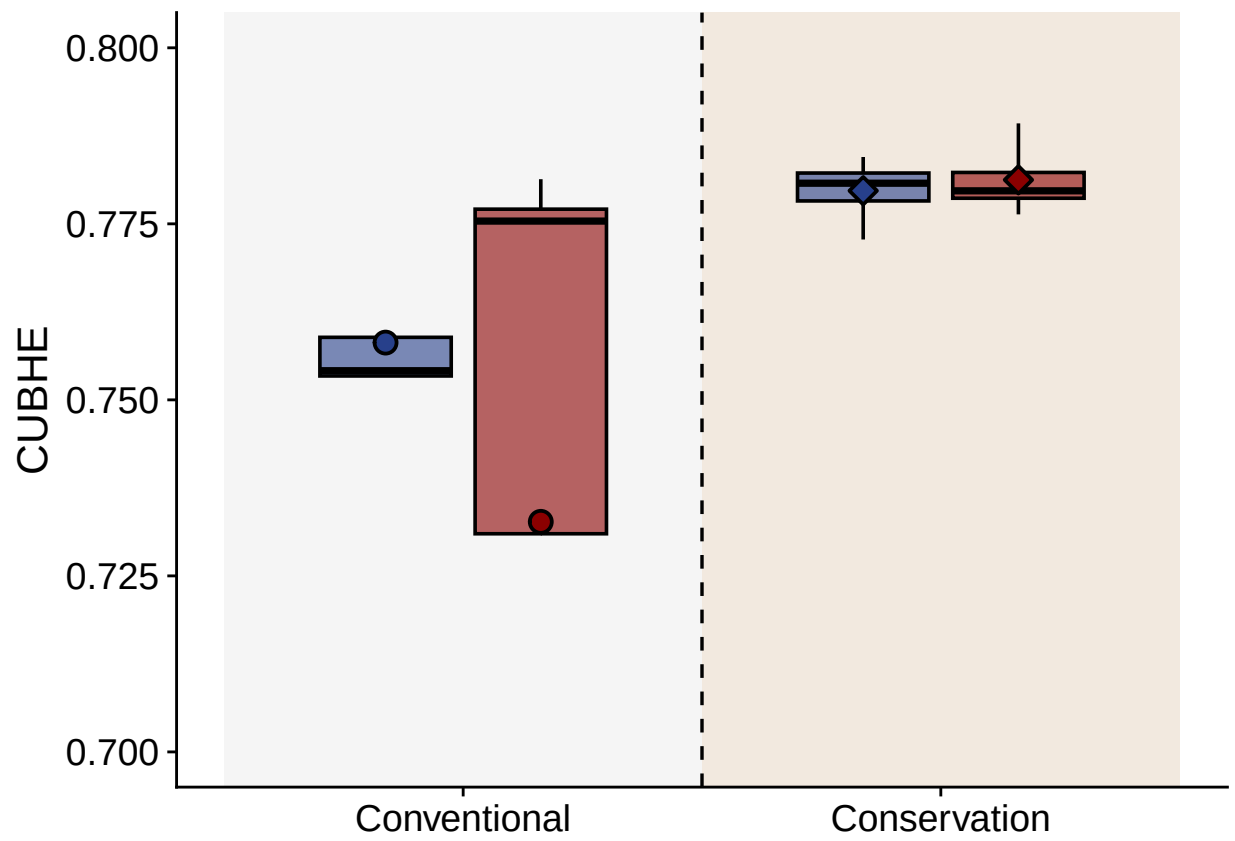
theme(axis.text = element_text(size = 14), axis.title.y = element_text(size = 16), legend.
      {if(v == "CUBHE") coord_cartesian(ylim = c(0.70, 0.8))} # CUBHE
}

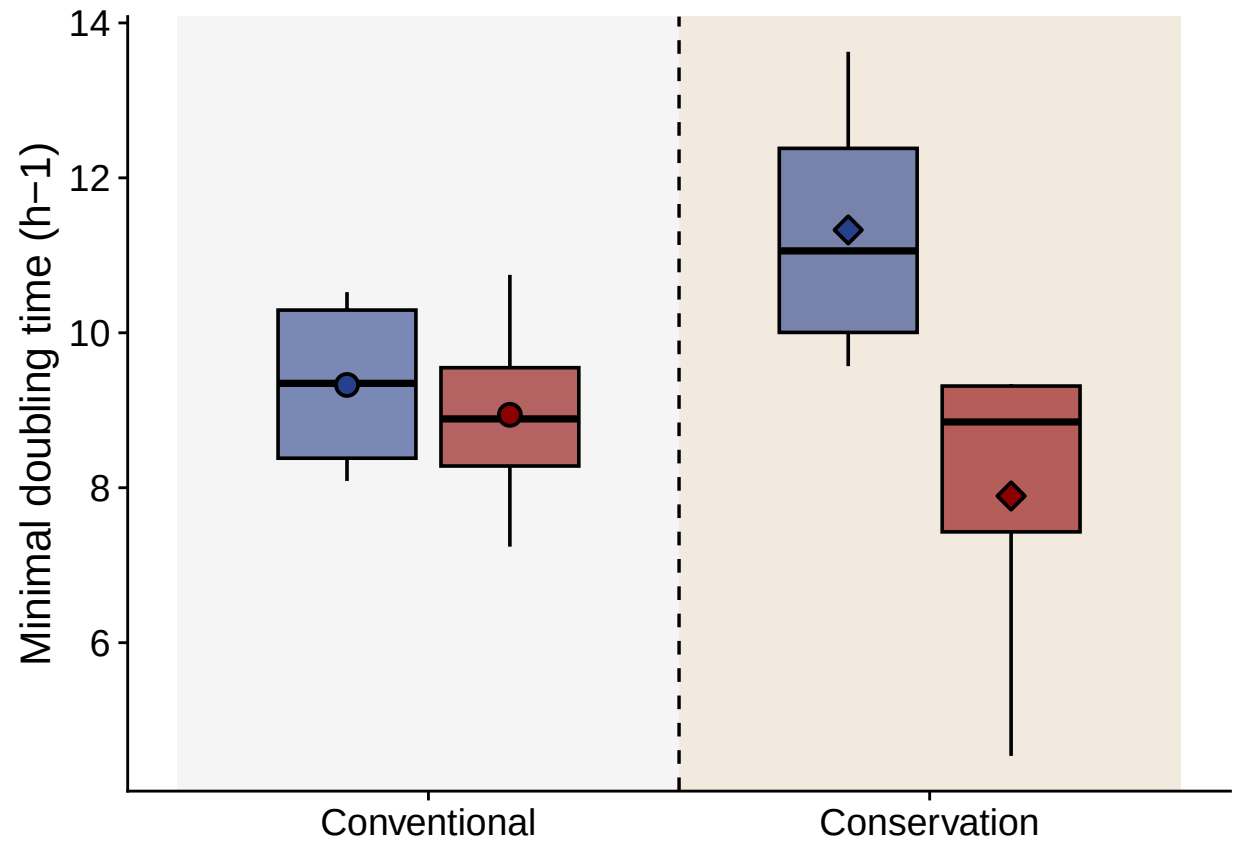
plots$AGS_Mb; plots$ACN; plots$CUBHE; plots$d

```

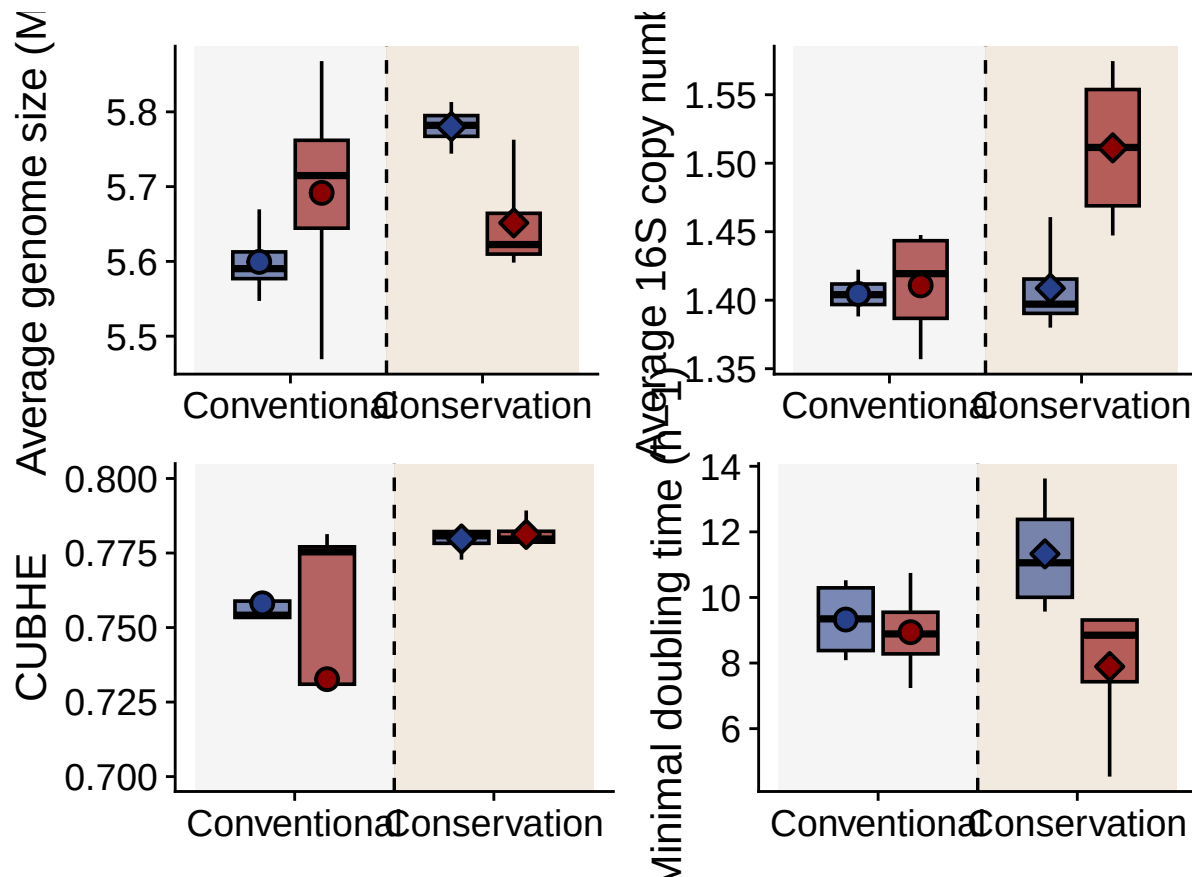








```
(plots$AGS_Mb | plots$ACN) /  
(plots$CUBHE | plots$d)
```

0.5.3 D. plot in <https://itol.embl.de/>

0.5.4 E. microbial life-history traits at MAGs level

```
MAG.traits.long_df <- read.xlsx("data_for_RAE.xlsx", sheet = 4) %>%
  group_by(ID, Treat) %>%
  mutate(
    Block = paste0("B", row_number()),
    Sample = paste0(Treat, row_number())
  ) %>%
  ungroup() %>%
  mutate(
    Management = factor(Management, c("Conventional", "Conservation")),
    Warm = factor(Warm, c("Ambient", "Warming")),
    Treat = factor(Treat, c("TN", "TW", "NN", "NW")),
    Block = factor(Block)
  ); head(MAG.traits.long_df )
```

```
## # A tibble: 6 x 10
```

```
##   ID      Treat Management    Warm  abundance CUBHE      d tRNA_count Block Sample
##   <chr> <fct> <fct>      <fct>    <dbl> <dbl> <dbl>      <dbl> <fct> <chr>
```

```
## 1 MAG1 NN Conservation Ambient 11.6 0.563 0.143 16 B1 NN1
## 2 MAG1 NN Conservation Ambient 3.08 0.563 0.143 16 B2 NN2
## 3 MAG1 NN Conservation Ambient 1.84 0.563 0.143 16 B3 NN3
## 4 MAG1 NN Conservation Ambient 1.60 0.563 0.143 16 B4 NN4
## 5 MAG1 NW Conservation Warming 3.24 0.563 0.143 19 B1 NW1
## 6 MAG1 NW Conservation Warming 3.84 0.563 0.143 19 B2 NW2
```

```
head(MAG.traits.long_df)
```

```
## # A tibble: 6 x 10
```

```
##   ID      Treat Management      Warm      abundance CUBHE      d tRNA_count Block Sample
##   <chr> <fct> <fct>          <fct>          <dbl> <dbl> <dbl>          <dbl> <fct> <chr>
## 1 MAG1 NN      Conservation Ambient      11.6 0.563 0.143      16 B1 NN1
## 2 MAG1 NN      Conservation Ambient      3.08 0.563 0.143      16 B2 NN2
## 3 MAG1 NN      Conservation Ambient      1.84 0.563 0.143      16 B3 NN3
## 4 MAG1 NN      Conservation Ambient      1.60 0.563 0.143      16 B4 NN4
## 5 MAG1 NW      Conservation Warming      3.24 0.563 0.143      19 B1 NW1
## 6 MAG1 NW      Conservation Warming      3.84 0.563 0.143      19 B2 NW2
```

```
MAG.weighted.traits_df <- MAG.traits.long_df %>%
  group_by(Sample, Treat, Management, Warm, Block) %>%
  summarise(
    total_MAG_abundance = sum(abundance, na.rm = TRUE),
    CUBHE_weighted_mean = weighted.mean(CUBHE, abundance, na.rm = TRUE),
    Growth_proxy_weighted_mean = weighted.mean(d, abundance, na.rm = TRUE),
    tRNA_count_weighted_mean = weighted.mean(tRNA_count, abundance, na.rm = TRUE),
    tRNA_count_weighted_sum = sum(abundance * tRNA_count, na.rm = TRUE),
    .groups = "drop"
  )
head(MAG.weighted.traits_df)
```

```
## # A tibble: 6 x 10
```

```
##   Sample Treat Management      Warm      Block total_MAG_abundance CUBHE_weighted_mean
##   <chr> <fct> <fct>          <fct> <fct>          <dbl>          <dbl>
## 1 NN1 NN      Conservation Ambie~ B1              771.          0.529
## 2 NN2 NN      Conservation Ambie~ B2              713.          0.529
## 3 NN3 NN      Conservation Ambie~ B3              811.          0.526
## 4 NN4 NN      Conservation Ambie~ B4              708.          0.531
## 5 NW1 NW      Conservation Warmi~ B1              784.          0.533
## 6 NW2 NW      Conservation Warmi~ B2              806.          0.533
## # i 3 more variables: Growth_proxy_weighted_mean <dbl>,
## #   tRNA_count_weighted_mean <dbl>, tRNA_count_weighted_sum <dbl>
```

```
MAG.traits.plot_df <- MAG.traits.long_df %>%
  select(ID, Sample, Treat, Management, Warm, Block, abundance, CUBHE, d, tRNA_count) %>%
  pivot_longer(c(CUBHE, d, tRNA_count), names_to = "Trait", values_to = "Value") %>%
```

```
mutate(Trait = factor(Trait, levels = c("CUBHE", "d", "tRNA_count")))
head(MAG.traits.plot_df)
```

```
## # A tibble: 6 x 9
##   ID      Sample Treat Management Warm      Block abundance Trait      Value
##   <chr> <chr>   <fct> <fct>      <fct> <fct>      <dbl> <fct>      <dbl>
## 1 MAG1  NN1     NN     Conservation Ambient B1      11.6 CUBHE      0.563
## 2 MAG1  NN1     NN     Conservation Ambient B1      11.6 d          0.143
## 3 MAG1  NN1     NN     Conservation Ambient B1      11.6 tRNA_count 16
## 4 MAG1  NN2     NN     Conservation Ambient B2      3.08 CUBHE      0.563
## 5 MAG1  NN2     NN     Conservation Ambient B2      3.08 d          0.143
## 6 MAG1  NN2     NN     Conservation Ambient B2      3.08 tRNA_count 16
```

```
mean.line_df <- MAG.traits.plot_df %>%
  group_by(Management, Warm, Trait) %>%
  summarise(mean_value = weighted.mean(Value, abundance, na.rm = TRUE), .groups = "drop")
head(mean.line_df)
```

```
## # A tibble: 6 x 4
##   Management Warm      Trait      mean_value
##   <fct>      <fct>   <fct>      <dbl>
## 1 Conventional Ambient CUBHE      0.523
## 2 Conventional Ambient d        1.44
## 3 Conventional Ambient tRNA_count 27.2
## 4 Conventional Warming CUBHE      0.518
## 5 Conventional Warming d        1.41
## 6 Conventional Warming tRNA_count 26.8
```

```
trait.model_map_df <- tibble(
  Trait = factor(c("CUBHE", "d", "tRNA_count"), levels = c("CUBHE", "d", "tRNA_count")),
  model_var = c("CUBHE_weighted_mean", "Growth_proxy_weighted_mean", "tRNA_count_weighted_mean")
)
head(trait.model_map_df)
```

```
## # A tibble: 3 x 2
##   Trait      model_var
##   <fct>      <chr>
## 1 CUBHE      CUBHE_weighted_mean
## 2 d          Growth_proxy_weighted_mean
## 3 tRNA_count tRNA_count_weighted_mean
```

```
run_trait_lme <- function(model_var, trait_label) {
  mod <- lme(
    as.formula(paste0(model_var, " ~ Management * Warm")),
    random = ~1 | Block,
```

```

    data = MAG.weighted.traits_df,
    method = "ML"
  )

  anova_df <- anova(mod) %>%
    as.data.frame() %>%
    rownames_to_column("term") %>%
    mutate(Trait = trait_label, model_var = model_var)

  pair_df <- pairs(
    emmeans(mod, ~ Warm | Management),
    reverse = TRUE,
    adjust = "bonferroni"
  ) %>%
    as.data.frame() %>%
    mutate(Trait = trait_label, model_var = model_var)

  list(anova = anova_df, pairs = pair_df)
}

model.results <- pmap(
  list(trait.model_map_df$model_var, trait.model_map_df$Trait),
  run_trait_lme
)

trait.anova_df <- map_dfr(model.results, "anova")
trait.pairs_df <- map_dfr(model.results, "pairs")
head(trait.pairs_df)

```

```

##           contrast Management      estimate      SE df    t.ratio
## 1 Warming - Ambient Conventional -0.005327832 0.001281284  9 -4.1581986
## 2 Warming - Ambient Conservation  0.004265556 0.001281284  9  3.3291264
## 3 Warming - Ambient Conventional -0.029867624 0.048034974  9 -0.6217891
## 4 Warming - Ambient Conservation -0.247765918 0.048034974  9 -5.1580317
## 5 Warming - Ambient Conventional -0.381751825 0.491526719  9 -0.7766655
## 6 Warming - Ambient Conservation  1.473545040 0.491526719  9  2.9978941
##           p.value      Trait      model_var
## 1 0.0024542759      CUBHE CUBHE_weighted_mean
## 2 0.0088113293      CUBHE CUBHE_weighted_mean
## 3 0.5495050745           d Growth_proxy_weighted_mean
## 4 0.0005966659           d Growth_proxy_weighted_mean
## 5 0.4572819084 tRNA_count tRNA_count_weighted_mean
## 6 0.0150075268 tRNA_count tRNA_count_weighted_mean

```

```

pval.real_df <- trait.pairs_df %>%
  mutate(
    Trait = factor(Trait, levels = levels(MAG.traits.plot_df$Trait)),

```

```

Management = factor(Management, levels = c("Conventional", "Conservation")),
p_value = p.value,
p_label = case_when(
  p_value < 0.001 ~ "P < 0.001",
  TRUE ~ paste0("P = ", signif(p_value, 2))
)
) %>%
select(Trait, Management, p_value, p_label, estimate, SE, df, t.ratio)
head(pval.real_df)

```

```

##      Trait  Management      p_value    p_label      estimate      SE df
## 1    CUBHE Conventional 0.0024542759 P = 0.0025 -0.005327832 0.001281284  9
## 2    CUBHE Conservation 0.0088113293 P = 0.0088  0.004265556 0.001281284  9
## 3      d Conventional 0.5495050745   P = 0.55 -0.029867624 0.048034974  9
## 4      d Conservation 0.0005966659   P < 0.001 -0.247765918 0.048034974  9
## 5 tRNA_count Conventional 0.4572819084   P = 0.46 -0.381751825 0.491526719  9
## 6 tRNA_count Conservation 0.0150075268   P = 0.015  1.473545040 0.491526719  9
##      t.ratio
## 1 -4.1581986
## 2  3.3291264
## 3 -0.6217891
## 4 -5.1580317
## 5 -0.7766655
## 6  2.9978941

```

```

plot_one_trait_density <- function(trait_name, show_strip = FALSE) {
  dat_df <- MAG.traits.plot_df %>% filter(Trait == trait_name)
  mean_df <- mean.line_df %>% filter(Trait == trait_name)

  p <- ggplot(dat_df, aes(Value, fill = Warm, color = Warm)) +
    geom_rect(
      data = tibble(Management = factor("Conventional", levels = c("Conventional", "Conservation")),
      aes(xmin = -Inf, xmax = Inf, ymin = -Inf, ymax = Inf),
      fill = bg.cols["Conventional"],
      alpha = 0.18,
      inherit.aes = FALSE
    ) +
    geom_rect(
      data = tibble(Management = factor("Conservation", levels = c("Conventional", "Conservation")),
      aes(xmin = -Inf, xmax = Inf, ymin = -Inf, ymax = Inf),
      fill = bg.cols["Conservation"],
      alpha = 0.18,
      inherit.aes = FALSE
    ) +
    geom_density(aes(weight = abundance), alpha = 0.10, linewidth = 0.8, adjust = 1.15) +
    geom_vline(data = mean_df, aes(xintercept = mean_value, color = Warm), linetype = "dashed")
  facet_grid(. ~ Management, scales = "free_x") +

```

```

scale_fill_manual(values = cols1) +
scale_color_manual(values = cols1) +
scale_y_continuous(labels = scales::label_number(accuracy = 0.01)) +
labs(x = NULL, y = trait_name, fill = NULL, color = NULL) +
theme_cowplot() +
theme(
  axis.text = element_text(size = 9),
  axis.title.y = element_text(size = 11, face = "bold"),
  legend.position = "none",
  panel.spacing.x = unit(0.45, "lines")
)

if (show_strip) {
  p + theme(
    strip.background = element_rect(fill = "grey88", color = NA),
    strip.text = element_text(size = 11, face = "bold")
  )
} else {
  p + theme(strip.background = element_blank(), strip.text = element_blank())
}
}

plot_one_p <- function(trait_name, show_x = FALSE) {
  p_min <- 0.001
  p_max <- 1
  x_lab <- if (show_x) "p value" else NULL

  bg_p_df <- tibble(
    Management = factor(c("Conservation", "Conventional"), levels = c("Conservation", "Conventional")),
    ymin = c(0.5, 1.5),
    ymax = c(1.5, 2.5),
    fill_color = c(bg.cols["Conservation"], bg.cols["Conventional"])
  )

  p <- pval.real_df %>%
    filter(Trait == trait_name) %>%
    mutate(
      Management = factor(Management, levels = c("Conservation", "Conventional")),
      p_plot = pmin(pmax(p_value, p_min), p_max),
      p_label = case_when(
        p_value < 0.001 ~ "p < 0.001",
        TRUE ~ paste0("p = ", signif(p_value, 2))
      ),
      label_x = case_when(
        p_plot <= 0.003 ~ p_plot * 2.2,
        p_plot >= 0.5 ~ p_plot / 2.0,
        TRUE ~ p_plot * 1.35
      )
    )
}

```

```

    ),
    label_hjust = case_when(
      p_plot >= 0.5 ~ 1,
      TRUE ~ 0
    )
  ) %>%
  ggplot(aes(p_plot, Management, shape = Management)) +
  geom_rect(
    data = bg_p_df,
    aes(xmin = p_min, xmax = p_max, ymin = ymin, ymax = ymax),
    fill = bg_p_df$fill_color,
    alpha = 0.18,
    inherit.aes = FALSE
  ) +
  geom_vline(xintercept = 0.05, linetype = "dashed", color = "grey50", linewidth = 0.5) +
  geom_point(size = 3.3, color = "black", fill = "red4", stroke = 0.9) +
  geom_text(aes(x = label_x, label = p_label, hjust = label_hjust), size = 4.3, show.legend = FALSE) +
  scale_shape_manual(values = c("Conventional" = 21, "Conservation" = 23)) +
  scale_x_log10(
    limits = c(p_min, p_max),
    breaks = c(0.001, 0.01, 0.05, 0.10, 1),
    labels = c("0.001", "0.01", "0.05", "0.10", "1"),
    guide = guide_axis(check.overlap = TRUE)
  ) +
  labs(x = x_lab, y = NULL) +
  coord_cartesian(clip = "off") +
  theme_cowplot() +
  theme(
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank(),
    panel.grid = element_blank(),
    legend.position = "none",
    plot.margin = margin(5.5, 45, 5.5, 5.5)
  )
)

if (show_x) {
  p + theme(
    axis.text.x = element_text(size = 9),
    axis.ticks.x = element_line(),
    axis.title.x = element_text(size = 11)
  )
} else {
  p + theme(
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    axis.title.x = element_blank()
  )
}

```

```
}
}
```

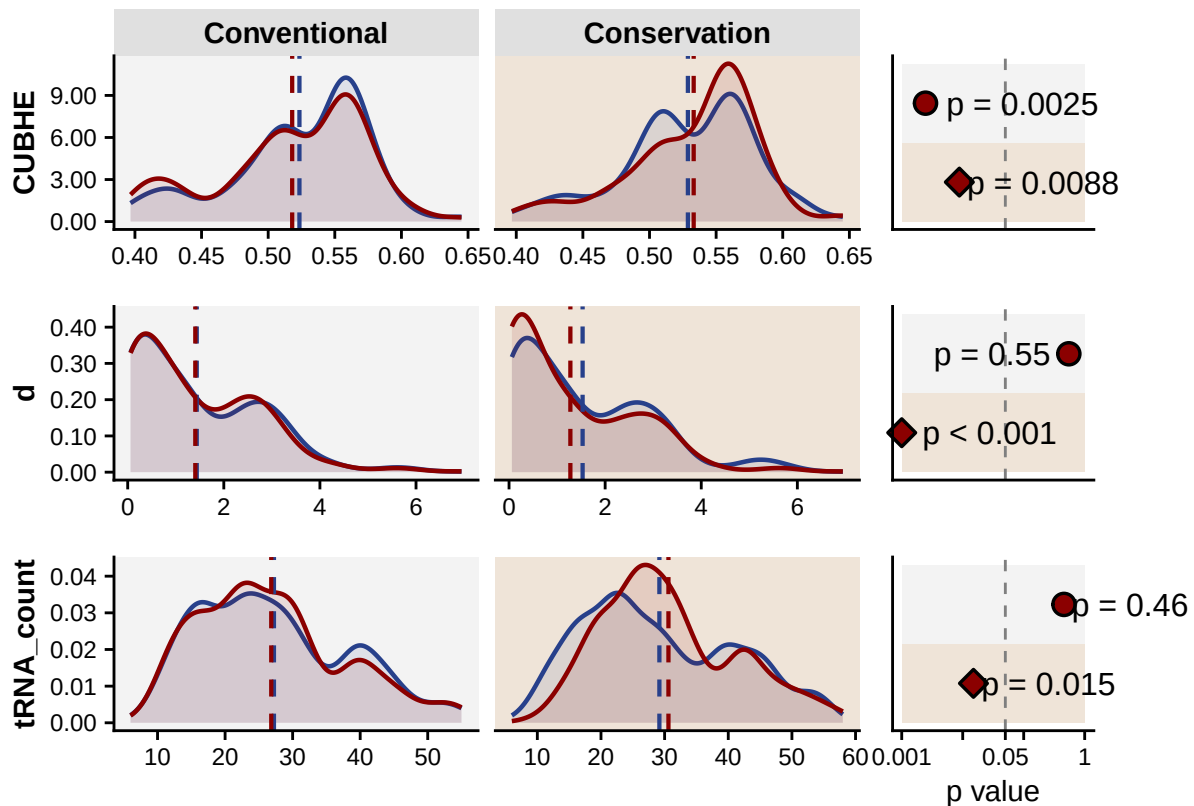
0.5.5 Figure assembly

```
trait_levels <- levels(MAG.traits.plot_df$Trait)

density_plots <- purrr::imap(trait_levels, ~ plot_one_trait_density(.x, show_strip = .y == 1))
p_plots <- purrr::imap(trait_levels, ~ plot_one_p(.x, show_x = .y == length(trait_levels)))

row_plots <- purrr::map2(
  density_plots,
  p_plots,
  ~ .x + .y + patchwork::plot_layout(widths = c(5, 1.35))
)

life_strategy_density_plot <- patchwork::wrap_plots(row_plots, ncol = 1)
life_strategy_density_plot
```



0.5.6 F. MAGs function

```
mags.df <- read.xlsx("data_for_RAE.xlsx", 5) %>%
  dplyr::select(Anno, System, Type, Function)
head(mags.df)
```

```
##      Anno System Type                Function
## 1  KEGG      NW GT39 Cell mebrane/wall components
## 2  KEGG      NW  GT4 Cell mebrane/wall components
## 3  KEGG      NW  GT4 Cell mebrane/wall components
## 4  KEGG      NW GT26 Cell mebrane/wall components
## 5 CAzyme      NW GH31                      Strach
## 6  KEGG      NW GT51                      murein synthesis
```

```
cazy.type_levels <- c(
  "CBM42", "CBM48",
  "GH5", "GH31", "GH29", "GH32",
  "GH13", "GH3", "GH51", "GH54", "GH15", "GH9",
  "GH20", "GH43"
)

NW.cazy_df <- mags.df %>%
  dplyr::filter(Anno == "CAzyme", System == "NW") %>%
  dplyr::distinct(Type, Anno, .keep_all = TRUE) %>%
  dplyr::select(Anno, System, Type)
head(NW.cazy_df)
```

```
##      Anno System Type
## 1 CAzyme      NW GH31
```

```
TW.cazy_df <- mags.df %>%
  dplyr::filter(Anno == "CAzyme", System == "TW") %>%
  dplyr::distinct(Type, Anno, .keep_all = TRUE) %>%
  dplyr::select(Anno, System, Type)
head(TW.cazy_df)
```

```
##      Anno System Type
## 1 CAzyme      TW  GH5
## 2 CAzyme      TW GH31
## 3 CAzyme      TW GH29
## 4 CAzyme      TW GH32
## 5 CAzyme      TW GH20
## 6 CAzyme      TW GH43
```

```
cazy.all_df <- dplyr::bind_rows(TW.cazy_df, NW.cazy_df)
head(cazy.all_df)
```

```
##      Anno System Type
## 1 CAzyme      TW  GH5
## 2 CAzyme      TW GH31
## 3 CAzyme      TW GH29
## 4 CAzyme      TW GH32
## 5 CAzyme      TW GH20
## 6 CAzyme      TW GH43
```

```
cazy.matrix_df <- cazy.all_df %>%
  dplyr::mutate(Present = 1) %>%
  dplyr::distinct(System, Type, .keep_all = TRUE) %>%
  tidyr::pivot_wider(names_from = System, values_from = Present, values_fill = 0)
head(cazy.matrix_df)
```

```
## # A tibble: 6 x 4
##   Anno   Type    TW    NW
##   <chr> <chr> <dbl> <dbl>
## 1 CAzyme GH5      1     0
## 2 CAzyme GH31     1     1
## 3 CAzyme GH29     1     0
## 4 CAzyme GH32     1     0
## 5 CAzyme GH20     1     0
## 6 CAzyme GH43     1     0
```

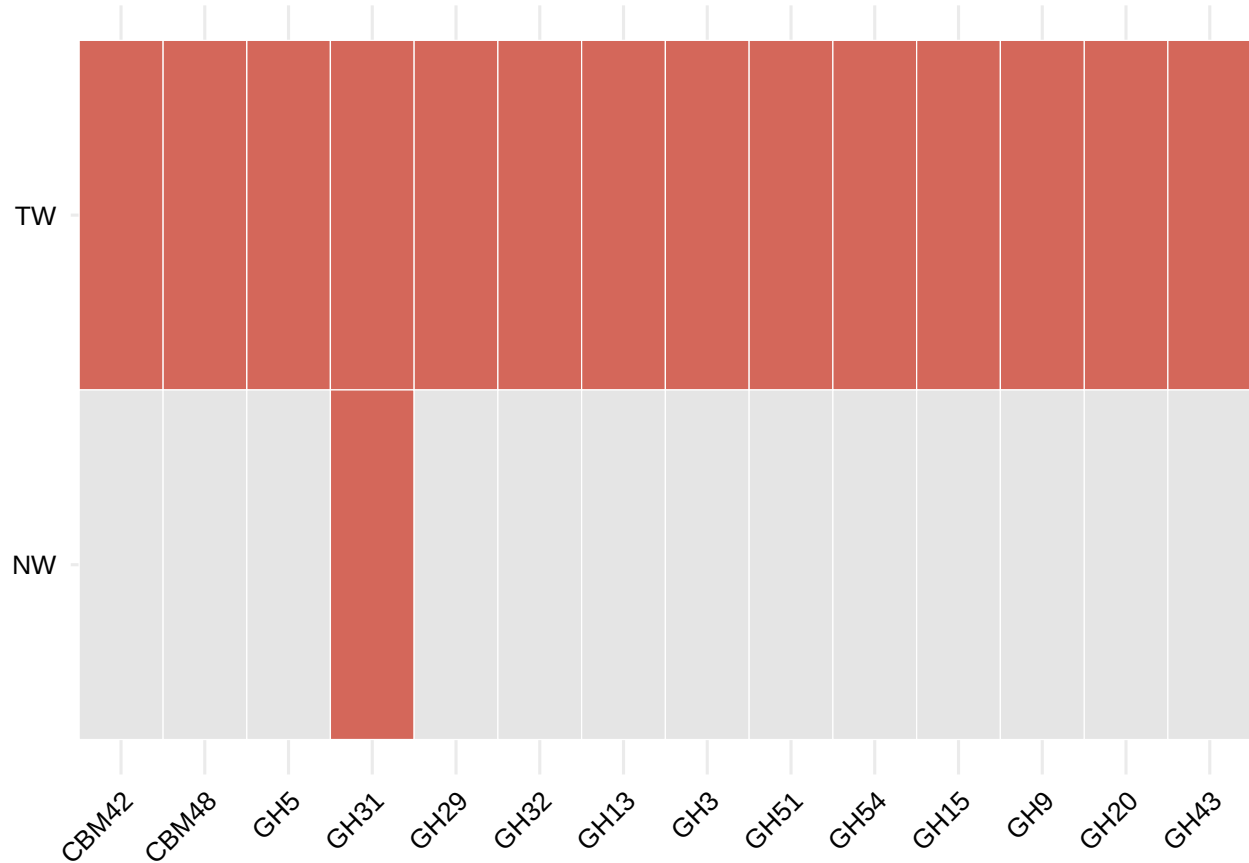
```
cazy.long_df <- cazy.matrix_df %>%
  dplyr::select(-Anno) %>%
  tidyr::pivot_longer(cols = -Type, names_to = "System", values_to = "Present") %>%
  dplyr::mutate(Type = factor(Type, levels = cazy.type_levels))
head(cazy.long_df)
```

```
## # A tibble: 6 x 3
##   Type System Present
##   <fct> <chr>    <dbl>
## 1 GH5   TW      1
## 2 GH5   NW      0
## 3 GH31  TW      1
## 4 GH31  NW      1
## 5 GH29  TW      1
## 6 GH29  NW      0
```

```

Cazyme.plot <- ggplot(cazy.long_df, aes(y = System, x = Type, fill = factor(Present))) +
  geom_tile(color = "white") +
  scale_fill_manual(
    values = c("0" = "grey90", "1" = "#d4675a"),
    name = "Presence",
    labels = c("Absent", "Present")
  ) +
  labs(x = NULL, y = NULL) +
  theme_minimal() +
  theme(
    axis.text = element_text(color = "black"),
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none"
  )
Cazyme.plot

```



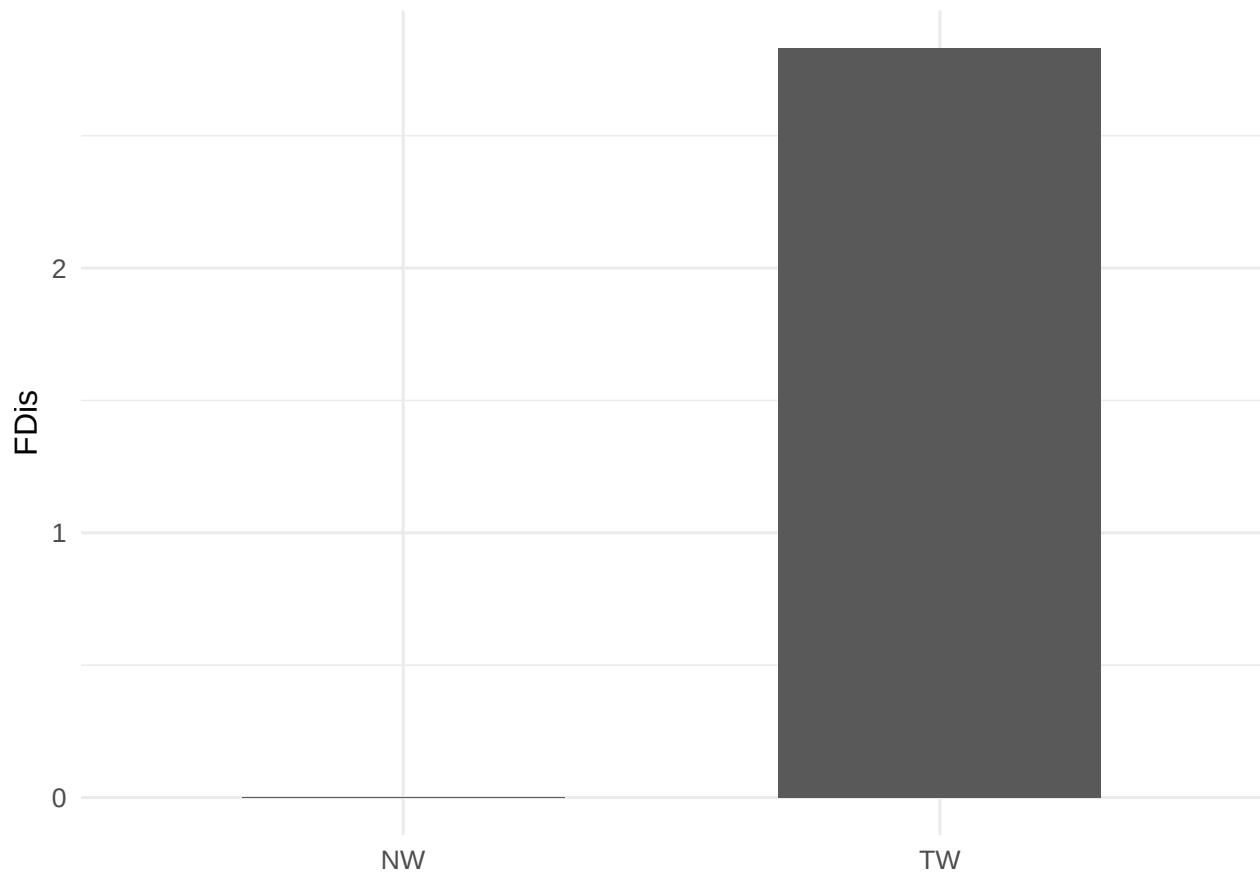
```

functi.df <- tibble(
  site = c("NW", "TW"),
  FDis = c(0.000000, 2.831448)
) # based on Functional Dispersion (FDis) from fundiversity package
head(functi.df)

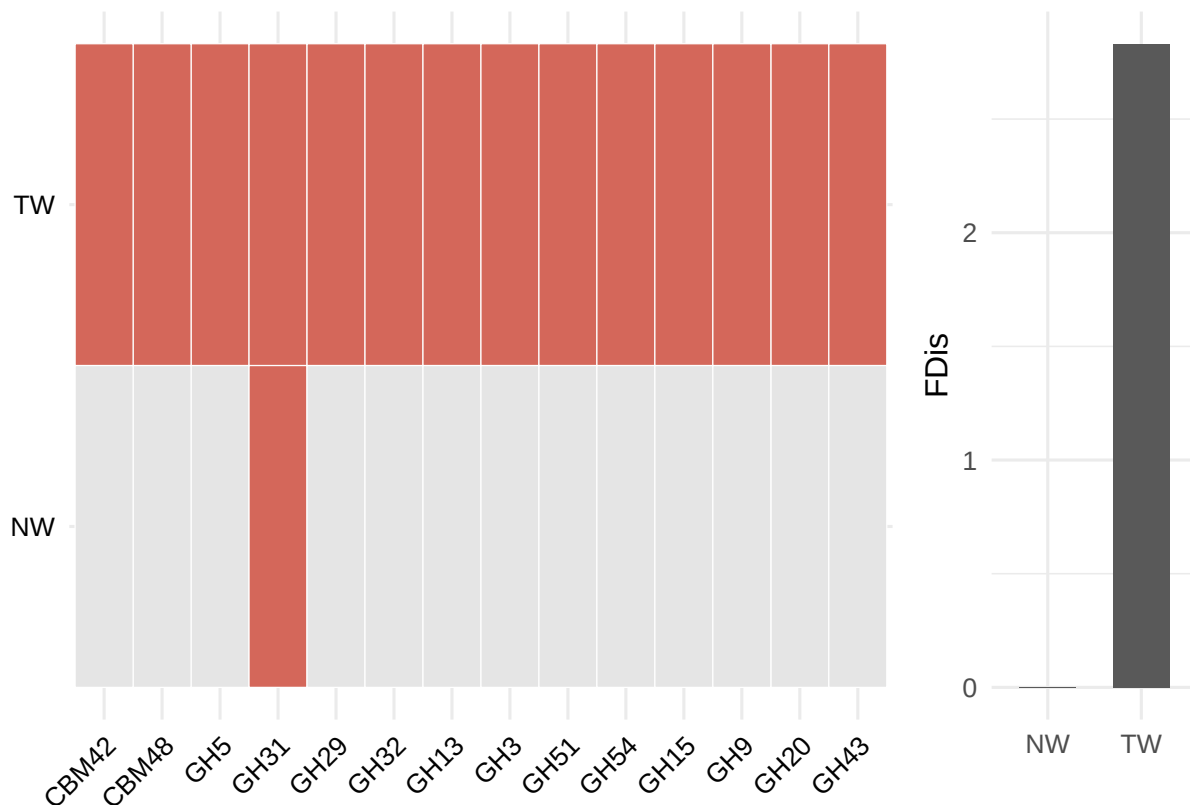
```

```
## # A tibble: 2 x 2
##   site  FDis
##   <chr> <dbl>
## 1 NW     0
## 2 TW    2.83
```

```
FDis.plot <- ggplot(funci.df, aes(site, FDis)) +
  geom_col(width = 0.6) +
  labs(x = NULL) +
  theme_minimal()
FDis.plot
```



```
cazy_plot <- (Cazyme.plot | FDis.plot) + patchwork::plot_layout(widths = c(4, 1))
cazy_plot
```



```
kegg.type_levels <- c(
  "4.1.2.13", "3.1.3.11", "2.7.2.3", "1.2.1.12", "5.3.1.6",
  "6.3.4.3", "6.4.1.2", "6.4.1.3", "5.1.99.1", "5.4.99.2",
  "4.2.1.17", "2.3.1.9", "1.3.5.1", "1.3.5.4",
  "4.2.1.2", "1.1.1.35", "1.1.1.37", "2.7.9.2",
  "GT4", "GT2", "GT35", "GT20", "GT28", "GT39", "GT26", "GT51"
)
```

```
NW.kegg_df <- mags.df %>%
  dplyr::filter(Anno == "KEGG", System == "NW") %>%
  dplyr::distinct(Type, Anno, .keep_all = TRUE) %>%
  dplyr::select(Anno, System, Type)
head(NW.kegg_df)
```

```
##   Anno System Type
## 1 KEGG     NW GT39
## 2 KEGG     NW  GT4
## 3 KEGG     NW GT26
## 4 KEGG     NW GT51
## 5 KEGG     NW GT28
## 6 KEGG     NW GT20
```

```

TW.kegg_df <- mags.df %>%
  dplyr::filter(Anno == "KEGG", System == "TW") %>%
  dplyr::distinct(Type, Anno, .keep_all = TRUE) %>%
  dplyr::select(Anno, System, Type)
head(TW.kegg_df)

```

```

##   Anno System Type
## 1 KEGG      TW  GT4
## 2 KEGG      TW  GT2
## 3 KEGG      TW GT35
## 4 KEGG      TW GT20
## 5 KEGG      TW GT28
## 6 KEGG      TW GT39

```

```

kegg.all_df <- dplyr::bind_rows(TW.kegg_df, NW.kegg_df)
head(kegg.all_df)

```

```

##   Anno System Type
## 1 KEGG      TW  GT4
## 2 KEGG      TW  GT2
## 3 KEGG      TW GT35
## 4 KEGG      TW GT20
## 5 KEGG      TW GT28
## 6 KEGG      TW GT39

```

```

kegg.matrix_df <- kegg.all_df %>%
  dplyr::mutate(Present = 1) %>%
  dplyr::distinct(System, Type, .keep_all = TRUE) %>%
  tidyr::pivot_wider(names_from = System, values_from = Present, values_fill = 0)
head(kegg.matrix_df)

```

```

## # A tibble: 6 x 4
##   Anno  Type      TW      NW
##   <chr> <chr> <dbl> <dbl>
## 1 KEGG  GT4      1      1
## 2 KEGG  GT2      1      0
## 3 KEGG  GT35     1      0
## 4 KEGG  GT20     1      1
## 5 KEGG  GT28     1      1
## 6 KEGG  GT39     1      1

```

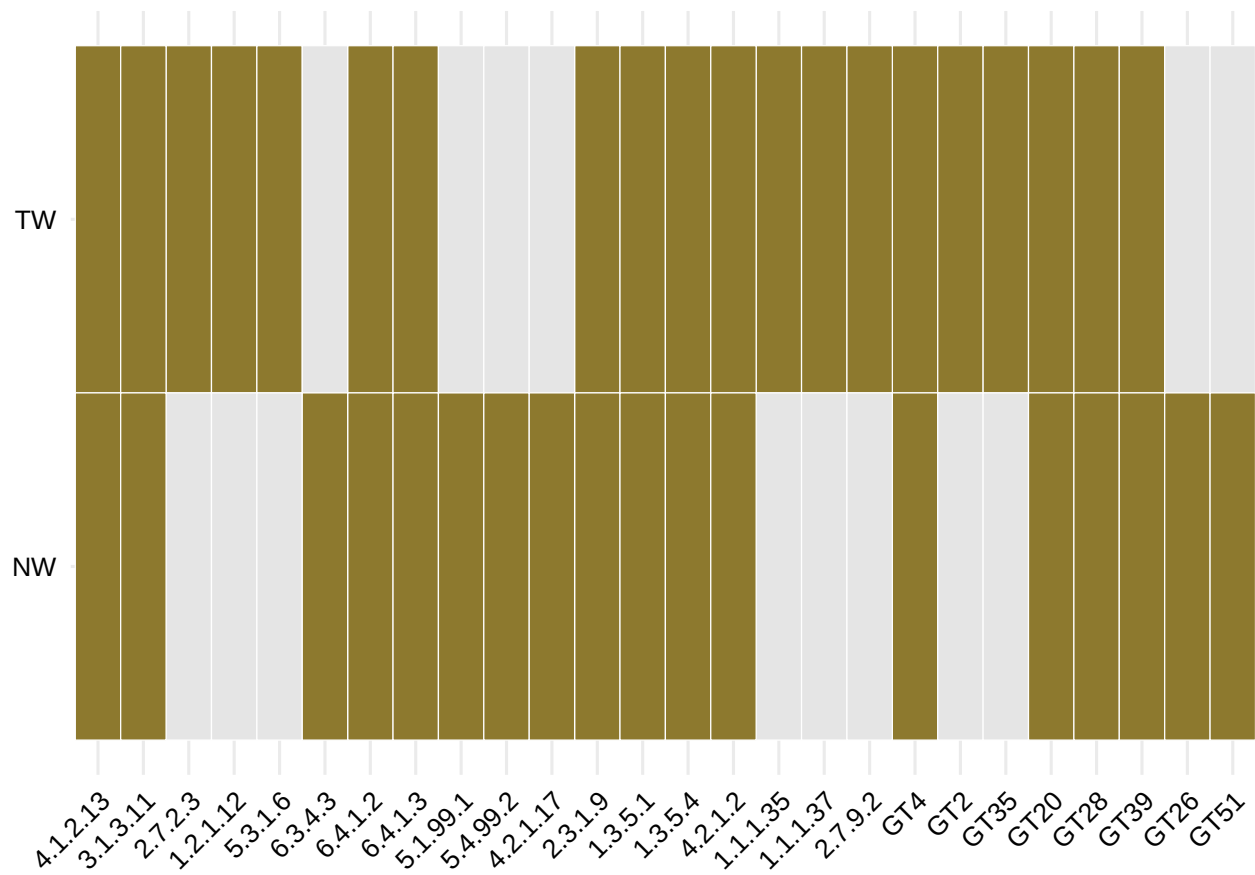
```

kegg.long_df <- kegg.matrix_df %>%
  dplyr::select(-Anno) %>%
  tidyr::pivot_longer(cols = -Type, names_to = "System", values_to = "Present") %>%
  dplyr::mutate(Type = factor(Type, levels = kegg.type_levels))
head(kegg.long_df)

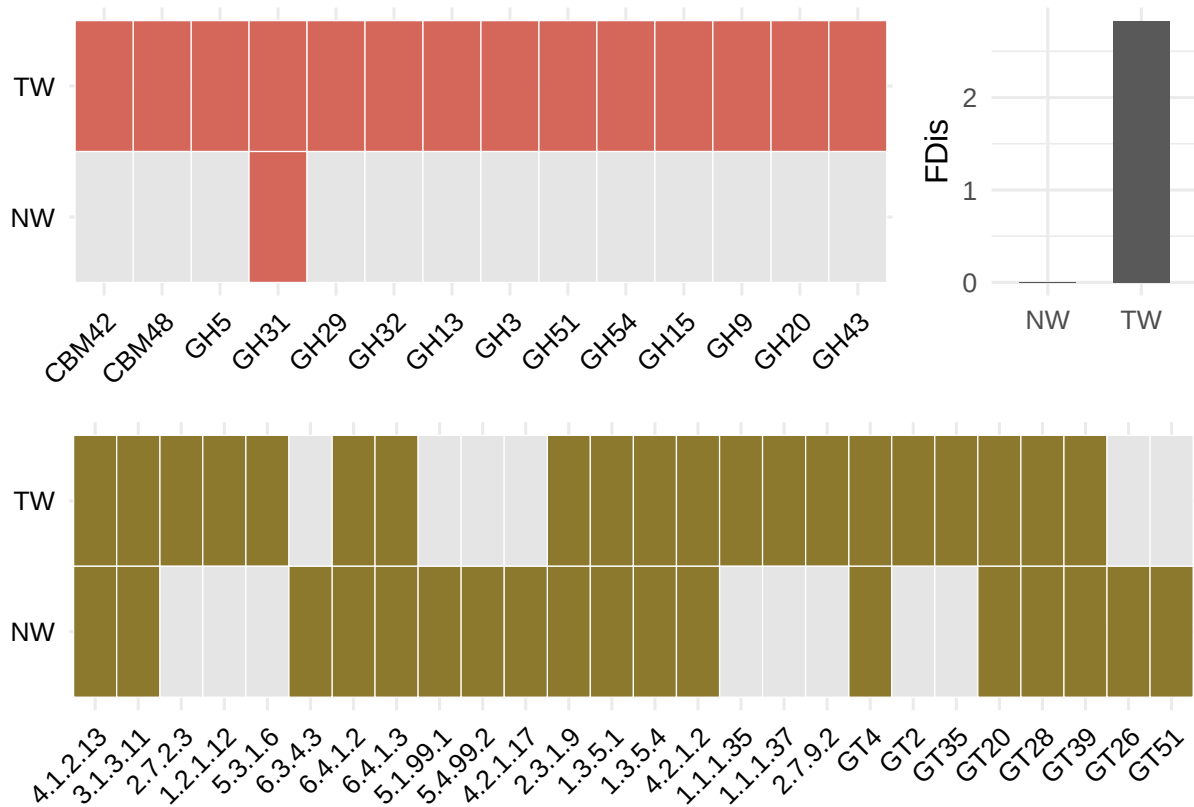
```

```
## # A tibble: 6 x 3
##   Type System Present
##   <fct> <chr>    <dbl>
## 1 GT4    TW         1
## 2 GT4    NW         1
## 3 GT2    TW         1
## 4 GT2    NW         0
## 5 GT35   TW         1
## 6 GT35   NW         0
```

```
kegg.plot <- ggplot(kegg.long_df, aes(y = System, x = Type, fill = factor(Present))) +
  geom_tile(color = "white") +
  scale_fill_manual(
    values = c("0" = "grey90", "1" = "#8d792e"),
    name = "Presence",
    labels = c("Absent", "Present")
  ) +
  labs(x = NULL, y = NULL) +
  theme_minimal() +
  theme(
    axis.text = element_text(color = "black"),
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none"
  )
kegg.plot
```



```
MAG.function_plot <- cazy_plot / kegg.plot
MAG.function_plot
```

0.6 Fig5 SEM analysis

```
df <- read.xlsx("data_for_RAE.xlsx", 6) %>%
  mutate(
    Temperature_scale = scale(Temperature),
    Substrate_quality = -1 * Substrate_quality,
    Substrate_available_scale = Substrate_available^2
  )

model_CUE <- psem(
  lm(CUE ~ Temperature_scale + RAE + Substrate_available + A_C_ratio + RAE, data = df),
  lm(RAE ~ Microbial_life + Substrate_quality + A_C_ratio, data = df),
  lm(A_C_ratio ~ Substrate_quality + Substrate_available, data = df),
  lm(Microbial_life ~ Till_m + Substrate_available + Substrate_quality, data = df),
  lm(Substrate_available ~ Till_m + Temperature_scale, data = df),
  lm(Substrate_quality ~ Till_m + Temperature_scale + Till_m:Temperature_scale, data = df)
)

model_CUE_summary <- summary(model_CUE)
```

|

|

```

model_CUE_fisherC <- fisherC(model_CUE)
model_CUE_AIC <- AIC(model_CUE, AIC.type = "loglik")
model_CUE_coefs <- coefs(model_CUE)
model_CUE_path_plot <- plot(model_CUE)

model_CUE_boot <- bootEff(
  model_CUE,
  R = 9999,
  catch.err = FALSE,
  parallel = "no",
  ncpus = 1,
  seed = 23
)

model_CUE_boot_eff <- semEff(model_CUE_boot)
model_CUE_boot_summary <- summary(model_CUE_boot_eff)

```

##		Effect	Bias	Std. Err.	Lower CI
##		-----	-----	-----	-----
##	DIRECT: Temperature.scale	-0.102	0.024	0.152	-0.405
##	RAE	0.367	-0.023	0.165	0.027
##	A.C.ratio	0.184	-0.034	0.173	-0.075
##	Substrate.available	-0.361	0.022	0.161	-0.634
##					
##	INDIRECT: Temperature.scale	-0.128	-0.012	0.120	-0.345
##	Till.m	-0.014	-0.020	0.120	-0.244
##	Till.m:Temperature.scale	0.175	-0.036	0.065	0.085
##	A.C.ratio	0.148	-0.021	0.074	0.040
##	Microbial.life	0.109	-0.020	0.063	0.009
##	Substrate.available	0.212	-0.043	0.077	0.116
##	Substrate.quality	0.261	-0.039	0.097	0.117
##					
##	TOTAL: Temperature.scale	-0.230	0.012	0.146	-0.484
##	Till.m	-0.014	-0.020	0.120	-0.244
##	Till.m:Temperature.scale	0.175	-0.036	0.065	0.085
##	RAE	0.367	-0.023	0.165	0.027
##	A.C.ratio	0.332	-0.055	0.145	0.119
##	Microbial.life	0.109	-0.020	0.063	0.009
##	Substrate.available	-0.149	-0.021	0.154	-0.395
##	Substrate.quality	0.261	-0.039	0.097	0.117
##					
##	MEDIATORS: RAE	0.866	-0.145	0.372	0.325
##	A.C.ratio	0.646	-0.129	0.277	0.267
##	Microbial.life	0.257	-0.058	0.146	0.042
##	Substrate.available	-0.192	-0.025	0.191	-0.516
##	Substrate.quality	0.250	-0.047	0.102	0.099
##	Upper CI				

```

## -----
##      0.180 |
##      0.653 | *
##      0.680 |
##     -0.035 | *
##          |
##      0.136 |
##      0.220 |
##      0.337 | *
##      0.349 | *
##      0.260 | *
##      0.572 | *
##      0.507 | *
##          |
##      0.077 |
##      0.220 |
##      0.337 | *
##      0.653 | *
##      0.715 | *
##      0.260 | *
##      0.287 |
##      0.507 | *
##          |
##      1.818 | *
##      1.573 | *
##      0.679 | *
##      0.311 |
##      0.512 | *

```

##			Effect	Bias	Std. Err.	Lower CI
##			-----	-----	-----	-----
##	DIRECT:	A.C.ratio	0.403	-0.029	0.132	0.148
##		Microbial.life	0.297	-0.040	0.142	0.074
##		Substrate.quality	0.089	-0.014	0.121	-0.127
##						
##	INDIRECT:	Temperature.scale	0.223	-0.023	0.110	0.035
##		Till.m	0.294	-0.035	0.115	0.098
##		Till.m:Temperature.scale	0.318	-0.065	0.092	0.223
##		Substrate.available	0.352	-0.042	0.109	0.189
##		Substrate.quality	0.383	-0.055	0.117	0.221
##						
##	TOTAL:	Temperature.scale	0.223	-0.023	0.110	0.035
##		Till.m	0.294	-0.035	0.115	0.098
##		Till.m:Temperature.scale	0.318	-0.065	0.092	0.223
##		A.C.ratio	0.403	-0.029	0.132	0.148
##		Microbial.life	0.297	-0.040	0.142	0.074
##		Substrate.available	0.352	-0.042	0.109	0.189
##		Substrate.quality	0.473	-0.069	0.133	0.296

##						
##	MEDIATORS: A.C.ratio		0.784		-0.073	0.314 0.242
##	Microbial.life		0.700		-0.133	0.317 0.251
##	Substrate.available		0.451		-0.058	0.148 0.238
##	Substrate.quality		0.453		-0.074	0.167 0.204

Upper CI

##	0.643	*
##	0.573	*
##	0.345	
##		
##	0.479	*
##	0.548	*
##	0.543	*
##	0.625	*
##	0.682	*
##		
##	0.479	*
##	0.548	*
##	0.543	*
##	0.643	*
##	0.573	*
##	0.625	*
##	0.758	*
##		
##	1.484	*
##	1.639	*
##	0.844	*
##	0.866	*

##			Effect		Bias		Std. Err.		Lower CI
##			-----		-----		-----		-----
##	DIRECT:	Substrate.available	0.446		-0.021		0.175		0.029
##		Substrate.quality	0.474		-0.020		0.156		0.123
##									
##	INDIRECT:	Temperature.scale	0.291		-0.019		0.135		0.000
##		Till.m	0.417		-0.027		0.143		0.132
##		Till.m:Temperature.scale	0.318		-0.029		0.123		0.086
##									
##	TOTAL:	Temperature.scale	0.291		-0.019		0.135		0.000
##		Till.m	0.417		-0.027		0.143		0.132
##		Till.m:Temperature.scale	0.318		-0.029		0.123		0.086
##		Substrate.available	0.446		-0.021		0.175		0.029
##		Substrate.quality	0.474		-0.020		0.156		0.123
##									
##	MEDIATORS:	Substrate.available	0.572		-0.041		0.213		0.126
##		Substrate.quality	0.454		-0.034		0.189		0.133
##	Upper CI								

```

## -----
##      0.707 | *
##      0.704 | *
##      |
##      0.530 |
##      0.674 | *
##      0.565 | *
##      |
##      0.530 |
##      0.674 | *
##      0.565 | *
##      0.707 | *
##      0.704 | *
##      |
##      0.915 | *
##      0.910 | *

```

		Effect	Bias	Std. Err.	Lower CI
		-----	-----	-----	-----
## DIRECT:	Till.m	-0.237	-0.011	0.157	-0.548
##	Substrate.available	0.582	-0.042	0.155	0.348
##	Substrate.quality	0.649	-0.063	0.157	0.391
##					
## INDIRECT:	Temperature.scale	0.377	-0.023	0.161	0.046
##	Till.m	0.554	-0.057	0.170	0.271
##	Till.m:Temperature.scale	0.436	-0.060	0.139	0.223
##					
## TOTAL:	Temperature.scale	0.377	-0.023	0.161	0.046
##	Till.m	0.318	-0.069	0.195	-0.058
##	Till.m:Temperature.scale	0.436	-0.060	0.139	0.223
##	Substrate.available	0.582	-0.042	0.155	0.348
##	Substrate.quality	0.649	-0.063	0.157	0.391
##					
## MEDIATORS:	Substrate.available	0.745	-0.061	0.209	0.436
##	Substrate.quality	0.623	-0.078	0.216	0.265

```

## Upper CI

```

```

## -----
##      0.077 |
##      0.817 | *
##      0.867 | *
##      |
##      0.668 | *
##      0.851 | *
##      0.701 | *
##      |
##      0.668 | *
##      0.585 |
##      0.701 | *

```

```
##      0.817 | *
##      0.867 | *
##      |
##      1.112 | *
##      1.098 | *
```

```
##                                     Effect      Bias      Std. Err.      Lower CI      Upper CI
##                                     -----      -
## DIRECT:      Temperature.scale | 0.730 | -0.014 | 0.104 | 0.466 | 0.875
##              Till.m           | 0.552 | -0.008 | 0.135 | 0.243 | 0.770
##              |
## INDIRECT:    -                | -      | -      | -      | -      | -
##              |
## TOTAL:       Temperature.scale | 0.730 | -0.014 | 0.104 | 0.466 | 0.875
##              Till.m           | 0.552 | -0.008 | 0.135 | 0.243 | 0.770
##              |
## MEDIATORS:   -                | -      | -      | -      | -      | -
##
##
## | *
## | *
## |
## |
## |
## | *
## | *
## |
## |
```

```
##                                     Effect      Bias      Std. Err.      Lower CI
##                                     -----      -
## DIRECT:      Temperature.scale | -0.073 | 0.015 | 0.229 | -0.515
##              Till.m           | 0.360 | -0.022 | 0.197 | -0.117
##              Till.m:Temperature.scale | 0.672 | -0.040 | 0.167 | 0.273
##              |
## INDIRECT:    -                | -      | -      | -      | -
##              |
## TOTAL:       Temperature.scale | -0.073 | 0.015 | 0.229 | -0.515
##              Till.m           | 0.360 | -0.022 | 0.197 | -0.117
##              Till.m:Temperature.scale | 0.672 | -0.040 | 0.167 | 0.273
##              |
## MEDIATORS:   -                | -      | -      | -      | -
## Upper CI
## -----
##      0.381 |
##      0.650 |
##      0.882 | *
```

```
##          |
##          - |
##          |
##    0.381 |
##    0.650 |
##    0.882 | *
```

```
funGetEff <- function(Tp, Ef) {
  data.frame(effType = Tp, Effect = Ef, row.names = names(Ef)) %>%
    tibble::rownames_to_column(var = "Predictor")
}

get_effect_df <- function(response, filler_predictor = NULL, predictor_levels) {
  effRes <- semEff::getEff(model_CUE_boot_eff, response)
  dEf <- effRes[[response]]$Direct
  indEf <- effRes[[response]]$Indirect
  totalEf <- effRes[[response]]$Total

  combinedDf <- rbind(
    funGetEff("Direct", dEf),
    funGetEff("Indirect", indEf),
    funGetEff("Total", totalEf)
  ) %>%
    dplyr::filter(Predictor != "(Intercept)")

  if (!is.null(filler_predictor)) {
    combinedDf <- combinedDf %>%
      dplyr::bind_rows(
        tibble(
          Predictor = filler_predictor,
          effType = c("Direct", "Indirect", "Total"),
          Effect = 0
        )
      )
  }

  combinedDf %>%
    dplyr::mutate(Predictor = factor(Predictor, levels = predictor_levels))
}

plot_effect_df <- function(effect_df, show_legend = FALSE, show_x_text = TRUE) {
  bar_df <- effect_df %>%
    dplyr::filter(effType %in% c("Direct", "Indirect"))

  point_df <- effect_df %>%
```

```

dplyr::filter(effType == "Total")

p <- ggplot() +
  geom_col(data = bar_df, aes(x = Predictor, y = Effect, fill = effType), width = 0.7) +
  geom_point(data = point_df, aes(x = Predictor, y = Effect), size = 5, shape = 21, fill = "v") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  scale_fill_manual(values = c(Direct = "#4C72B0", Indirect = "#DD8452"), name = "Effect type") +
  scale_y_continuous(limits = c(-0.55, 0.55)) +
  labs(x = NULL, y = "Standardized effect size") +
  theme_cowplot(font_size = 20)

if (show_legend) {
  p <- p + theme(legend.position = c(0.1, 0.3))
} else {
  p <- p + theme(legend.position = "none")
}

if (show_x_text) {
  p + theme(axis.text.x = element_text(angle = 30, hjust = 1, vjust = 1))
} else {
  p + theme(axis.text.x = element_blank())
}
}

CUE.effect_df <- get_effect_df(
  response = "CUE",
  filler_predictor = "Microbial.life.R1",
  predictor_levels = c(
    "Till.m", "Temperature.scale", "Till.m:Temperature.scale",
    "Substrate.available", "Substrate.quality", "Microbial.life.R1",
    "A.C.ratio", "RAE"
  )
)
head(CUE.effect_df)

```

```

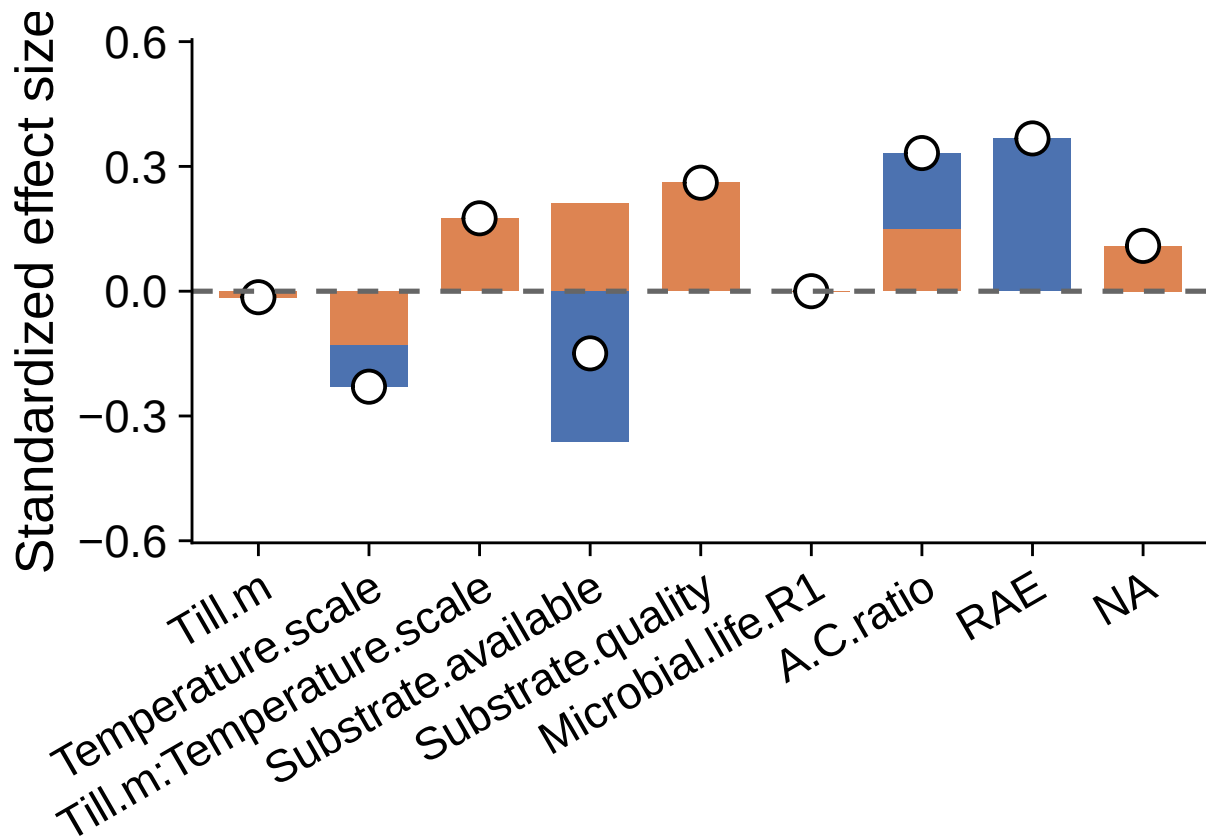
##           Predictor  effType      Effect
## 1 Temperature.scale  Direct -0.10155385
## 2              RAE    Direct  0.36730701
## 3       A.C.ratio    Direct  0.18415535
## 4 Substrate.available  Direct -0.36106319
## 5 Temperature.scale Indirect -0.12800937
## 6           Till.m Indirect -0.01446537

```

```

CUE_plot <- plot_effect_df(CUE.effect_df, show_legend = FALSE, show_x_text = TRUE)
CUE_plot

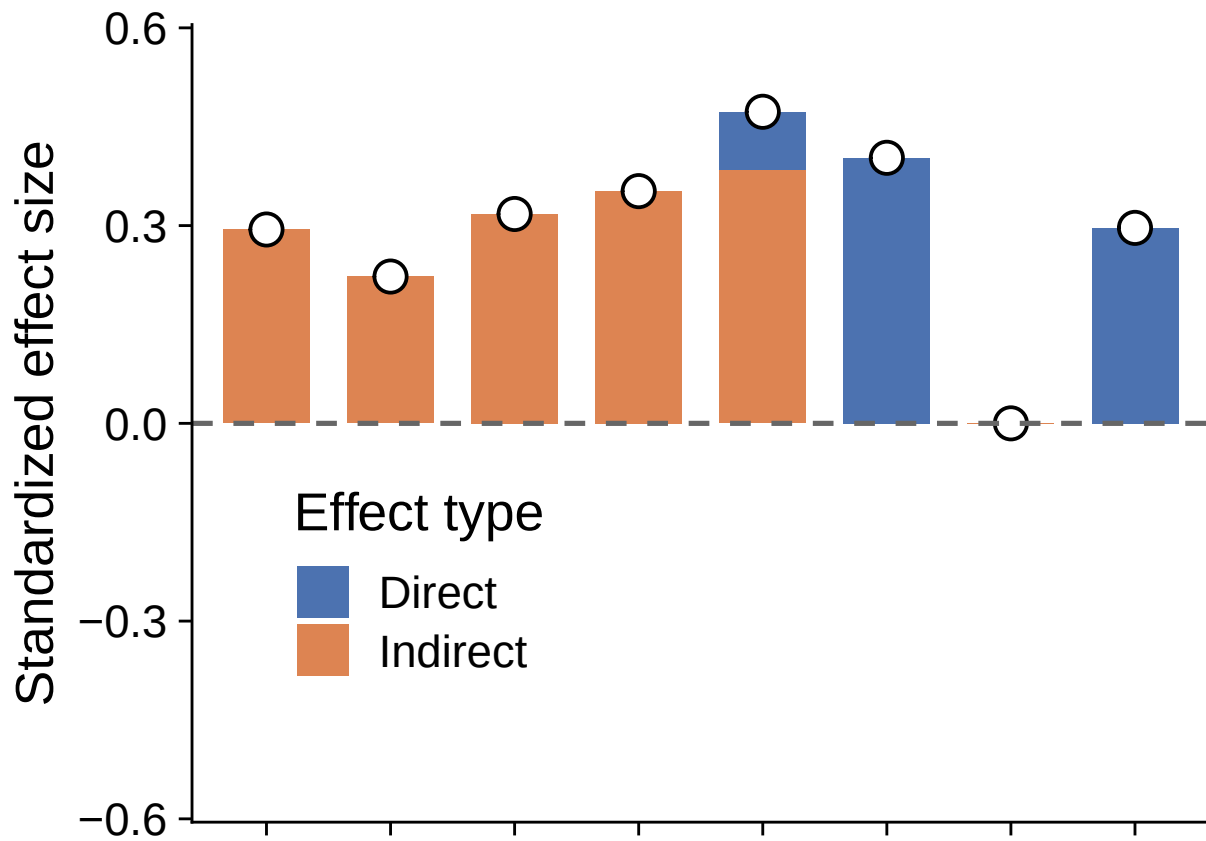
```

```
RAE.effect_df <- get_effect_df(
  response = "RAE",
  filler_predictor = "Perference",
  predictor_levels = c(
    "Till.m", "Temperature.scale", "Till.m:Temperature.scale",
    "Substrate.available", "Substrate.quality", "Microbial.life.R1",
    "A.C.ratio", "Perference"
  )
)
head(RAE.effect_df)
```

```
##           Predictor  effType    Effect
## 1           A.C.ratio   Direct 0.40287684
## 2              <NA>   Direct 0.29654852
## 3    Substrate.quality   Direct 0.08932739
## 4    Temperature.scale Indirect 0.22271192
## 5              Till.m Indirect 0.29410693
## 6 Till.m:Temperature.scale Indirect 0.31761412
```

```
RAE_plot <- plot_effect_df(RAE.effect_df, show_legend = TRUE, show_x_text = FALSE)
RAE_plot
```



```
SEM_effect_plot <- RAE_plot / CUE_plot
SEM_effect_plot
```

